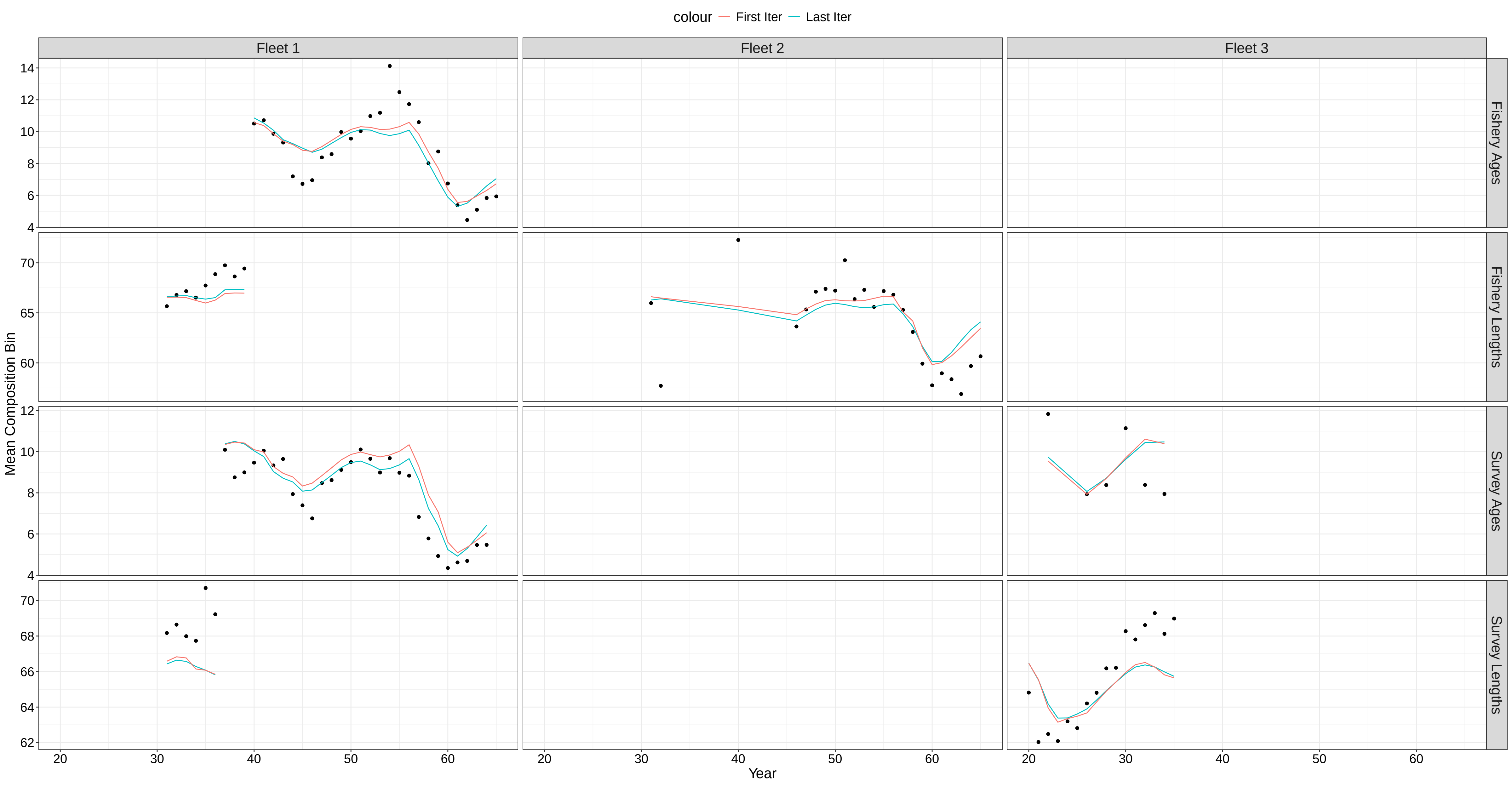
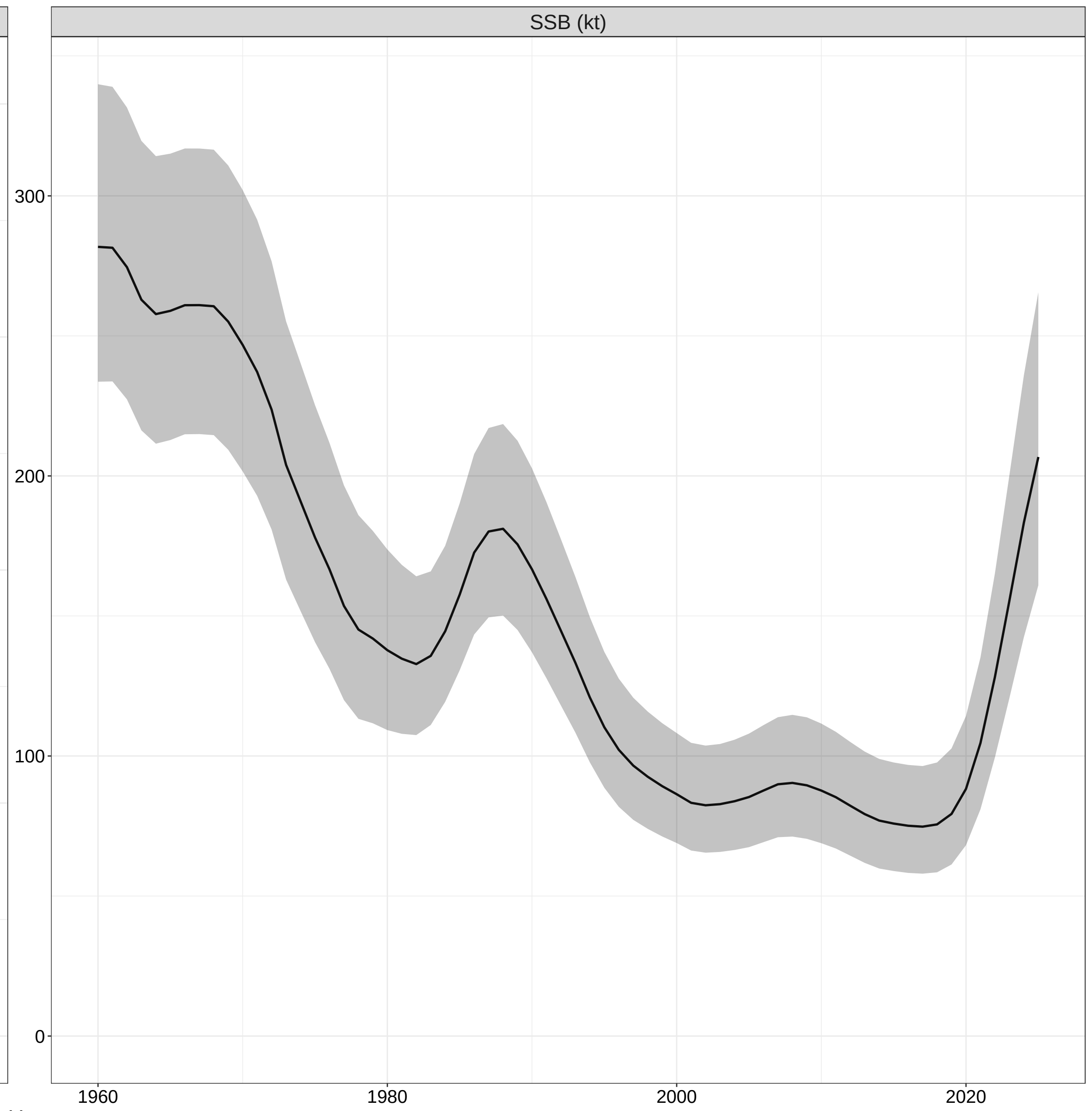
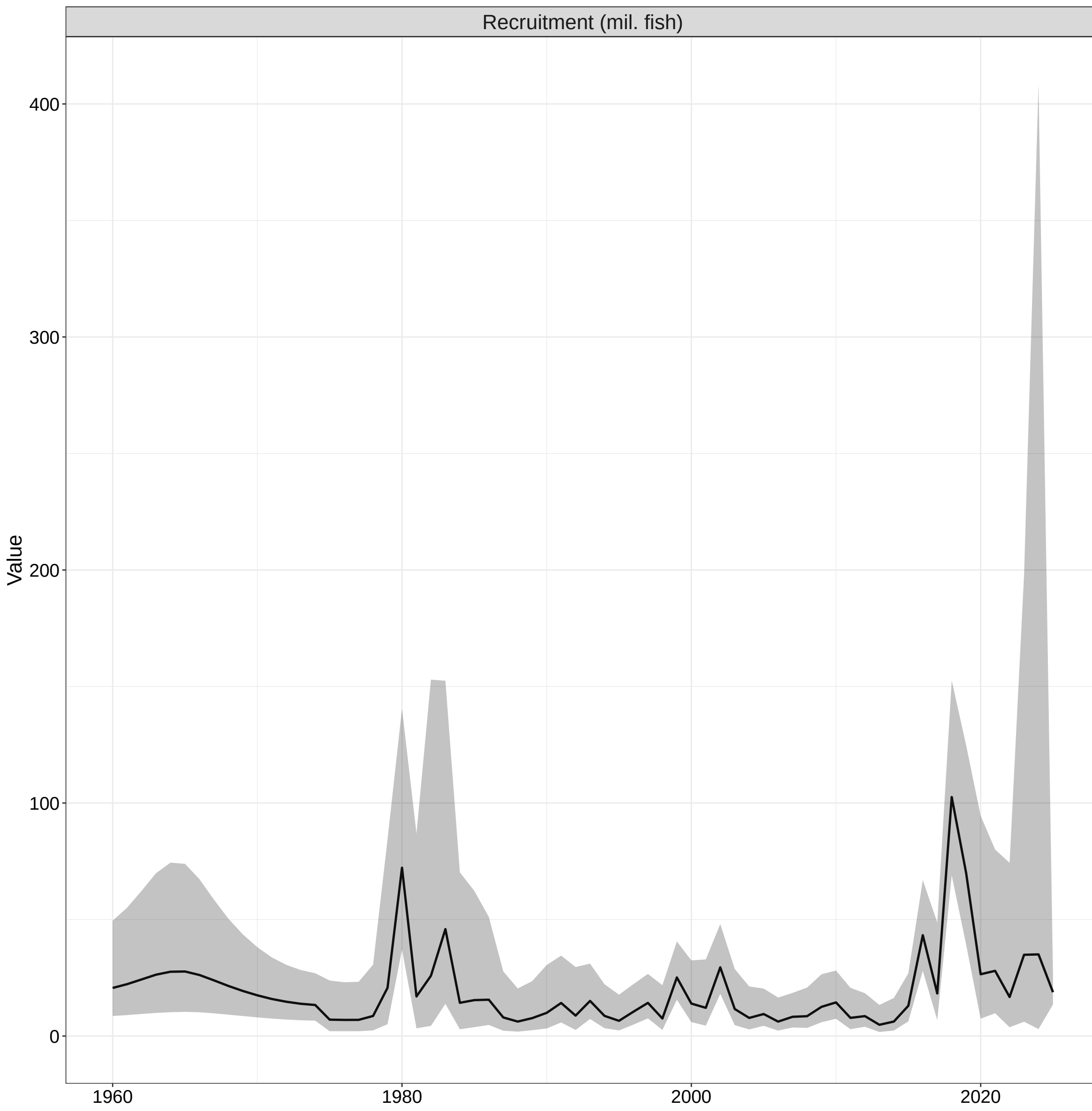
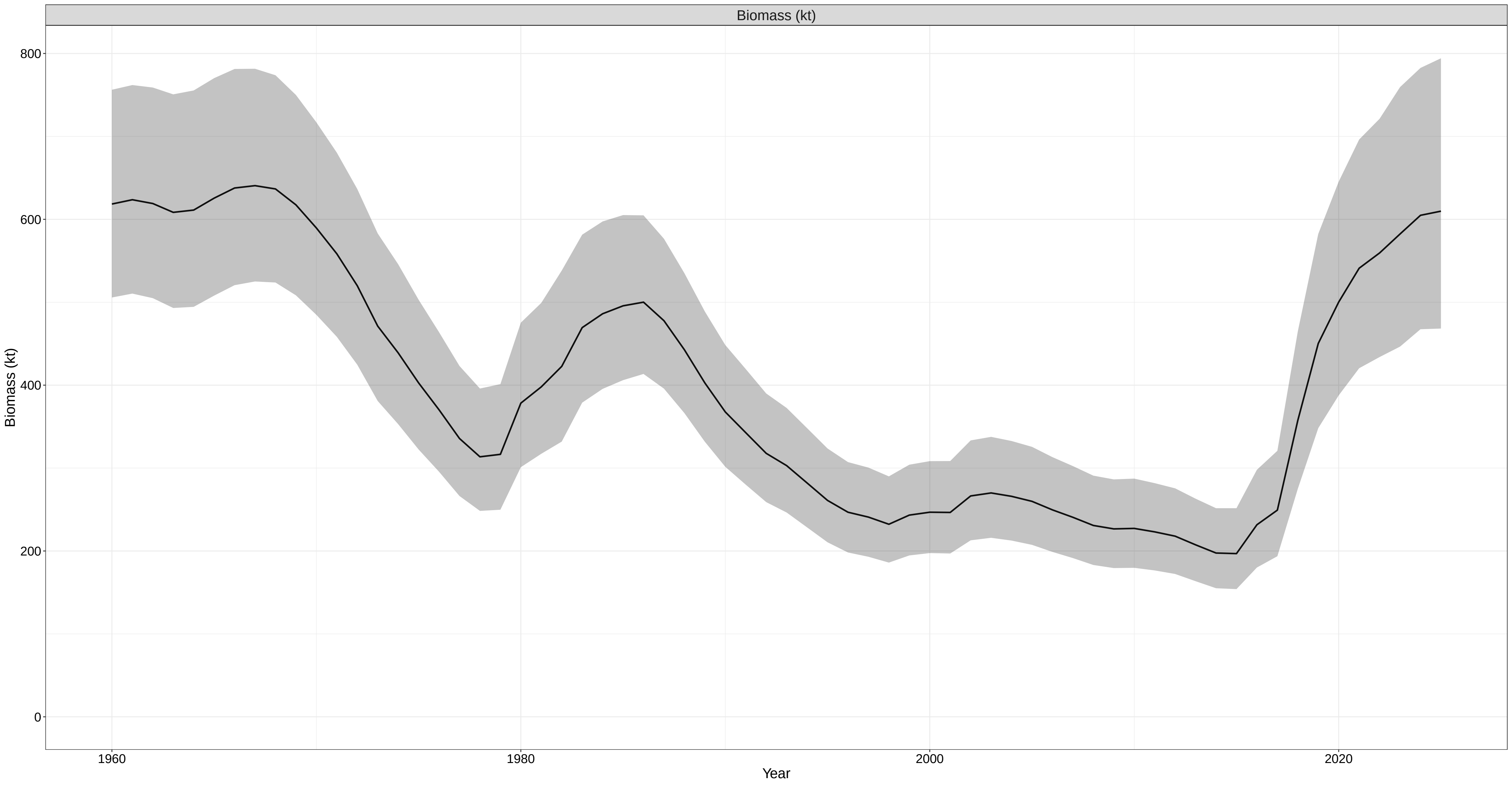


	Comp_Type	Init_Weight	Final_Weight
1	LLF_Age_F	0.8261073	0.7954410
2	LLF_Age_M	NA	NA
3	LLS_Age_F	3.7922454	3.5465392
4	LLS_Age_M	NA	NA
5	JPN_LLS_Age_F	1.3168111	0.6385227
6	JPN_LLS_Age_M	NA	NA
7	LLF_Len_F	4.1837057	2.8262846
8	LLF_Len_M	4.2696935	NA
9	TF_Len_F	0.3164859	0.3565194
10	TF_Len_M	0.2293966	NA
11	LLS_Len_F	1.4379202	2.3207368
12	LLS_Len_M	1.0705376	NA
13	JPN_LLS_Len_F	1.2777281	1.0527922
14	JPN_LLS_Len_M	0.8575195	NA



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	206.75434	358.76539	0.0487409092002144	39.08014	131.75946	0.07902	1.56918	0.57629	0.6168

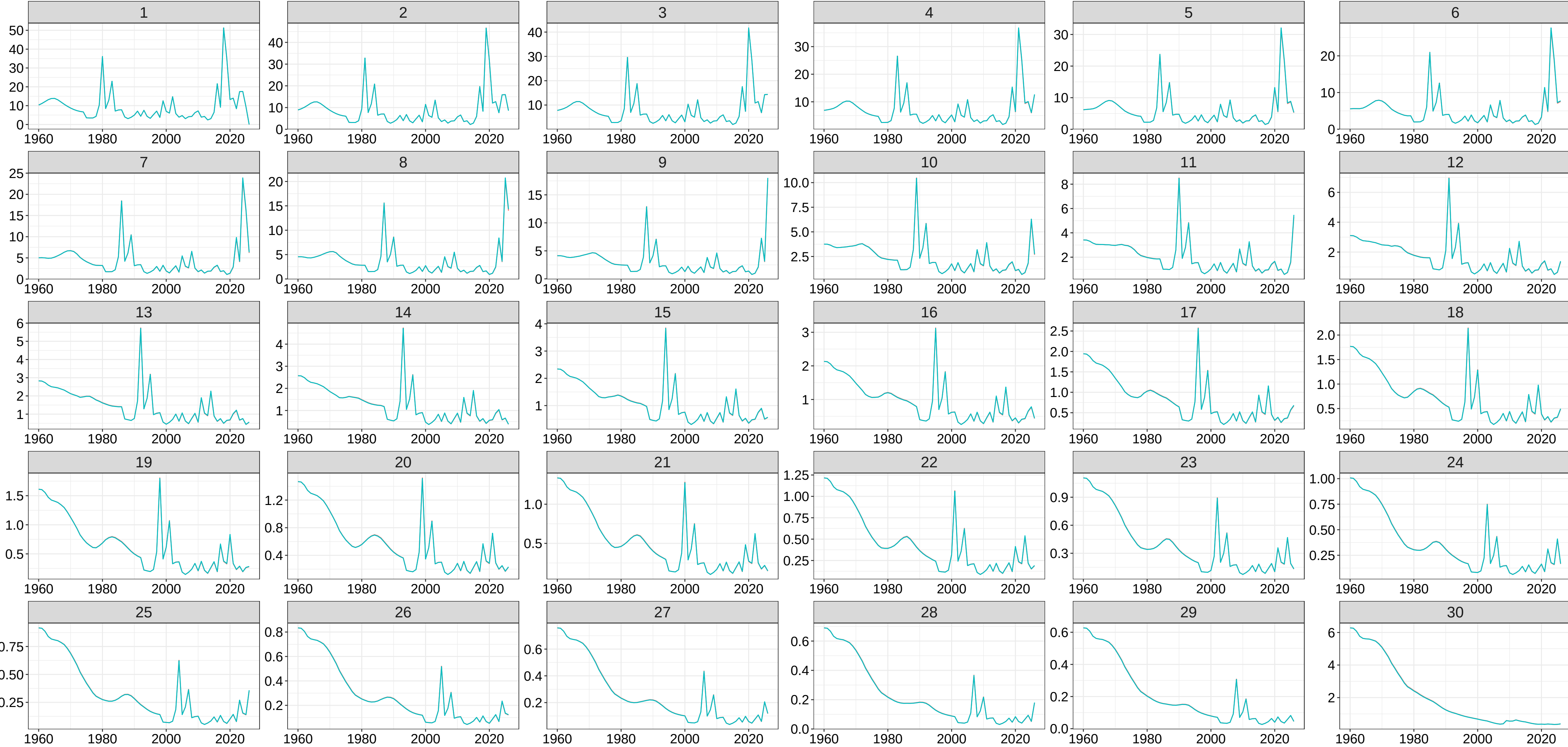


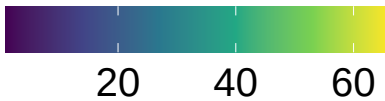


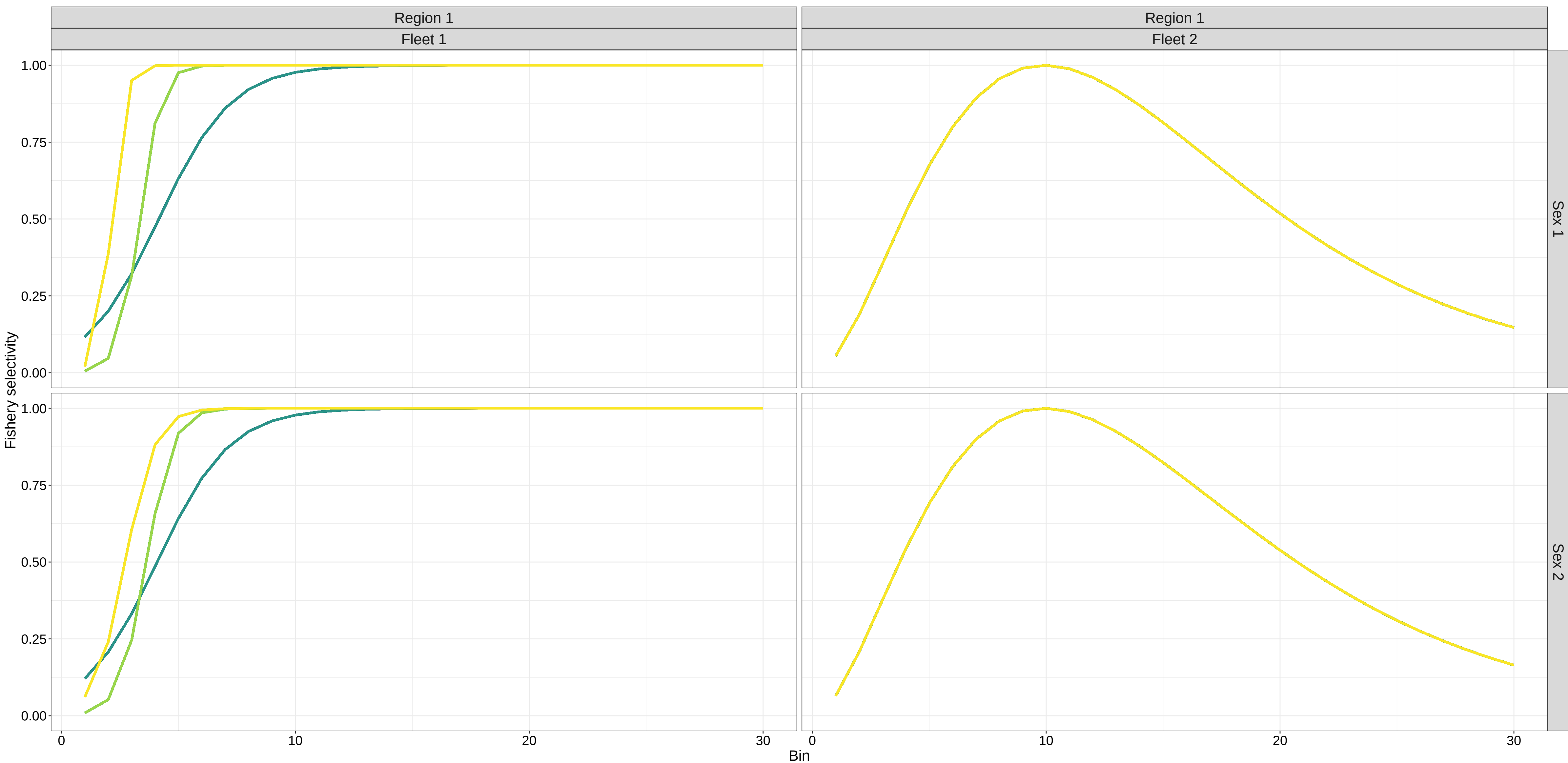
Numbers-at-Age

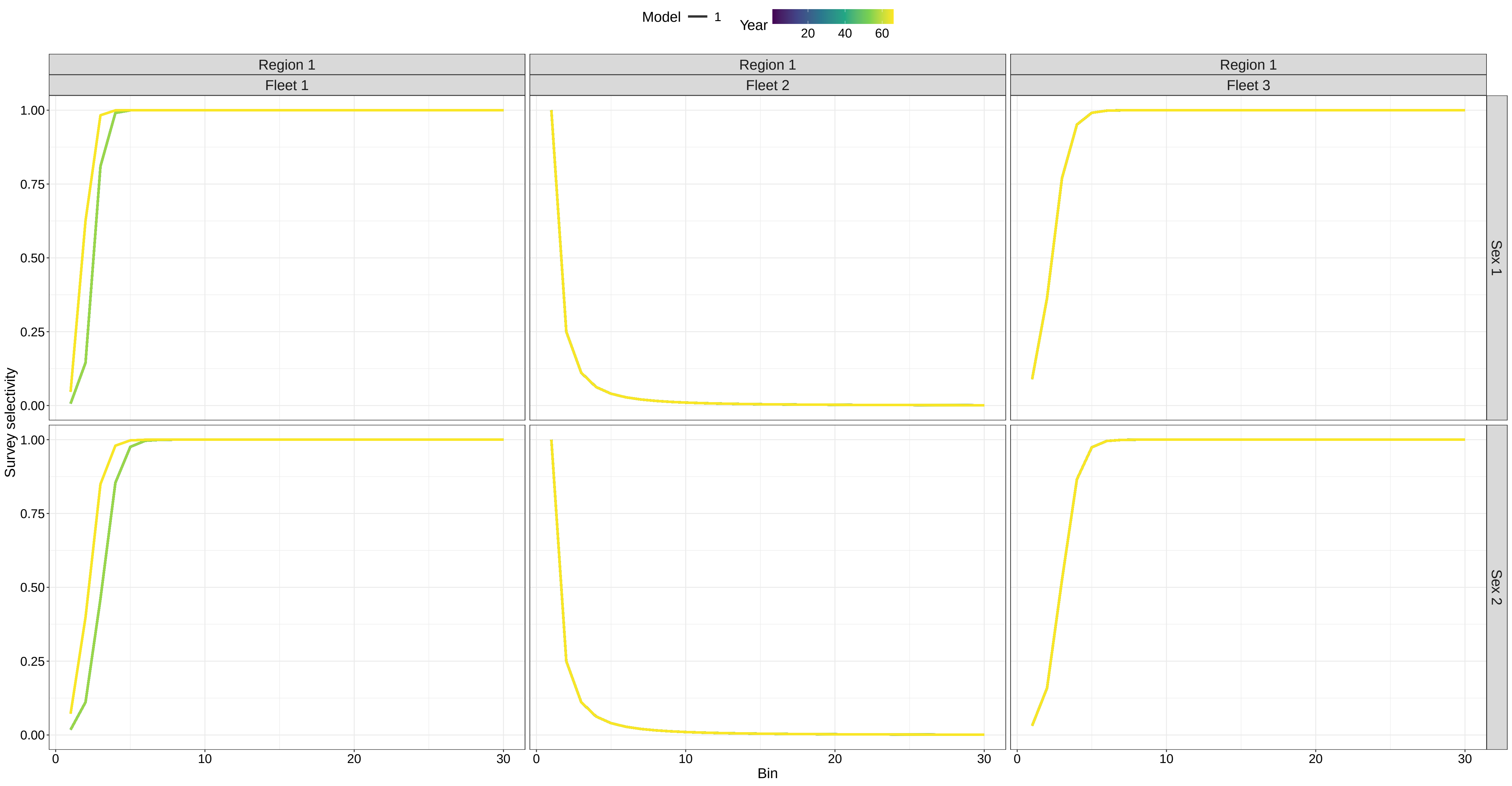
Sex — Female — Male

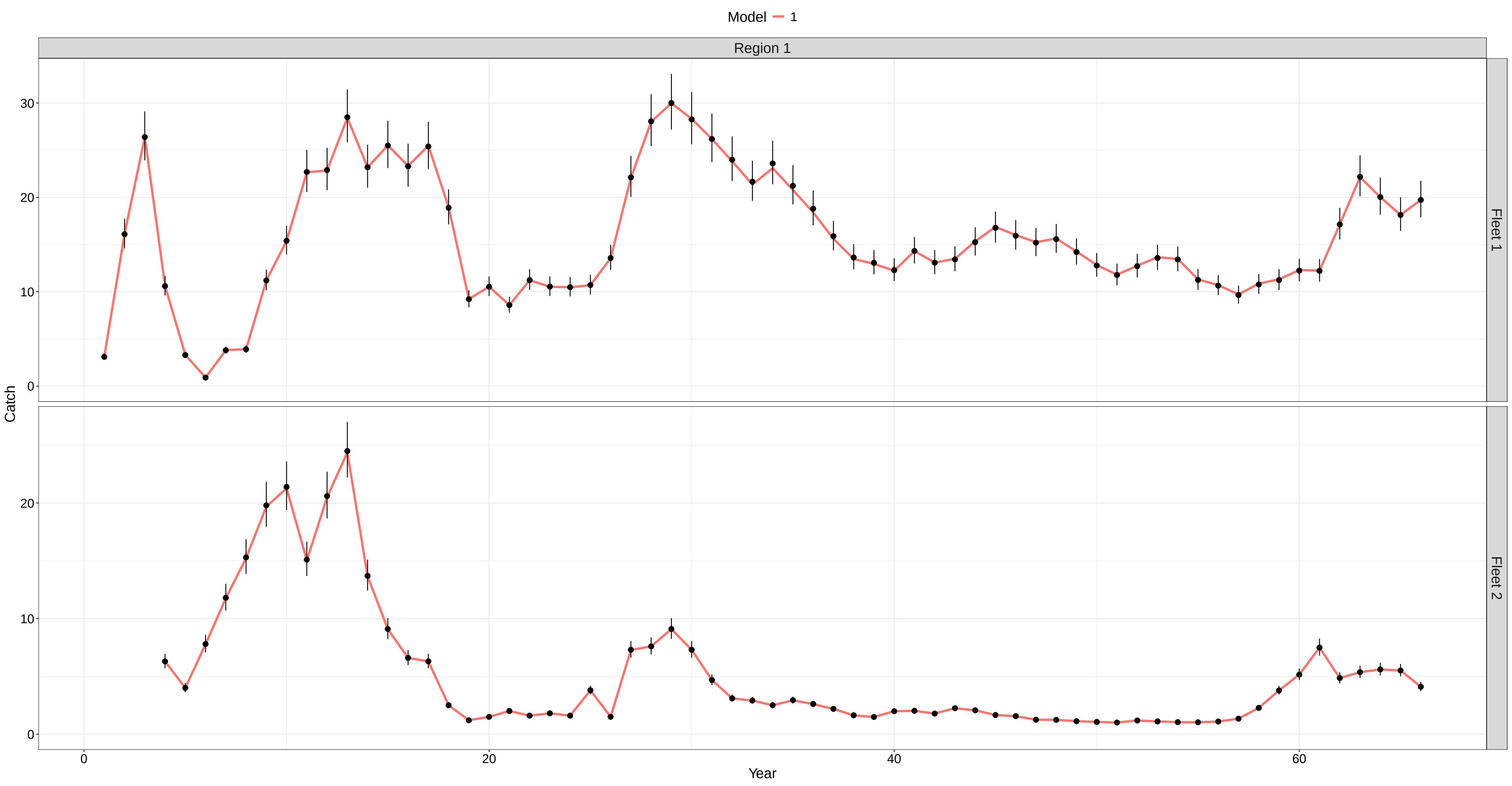
Numbers-at-age (millions)



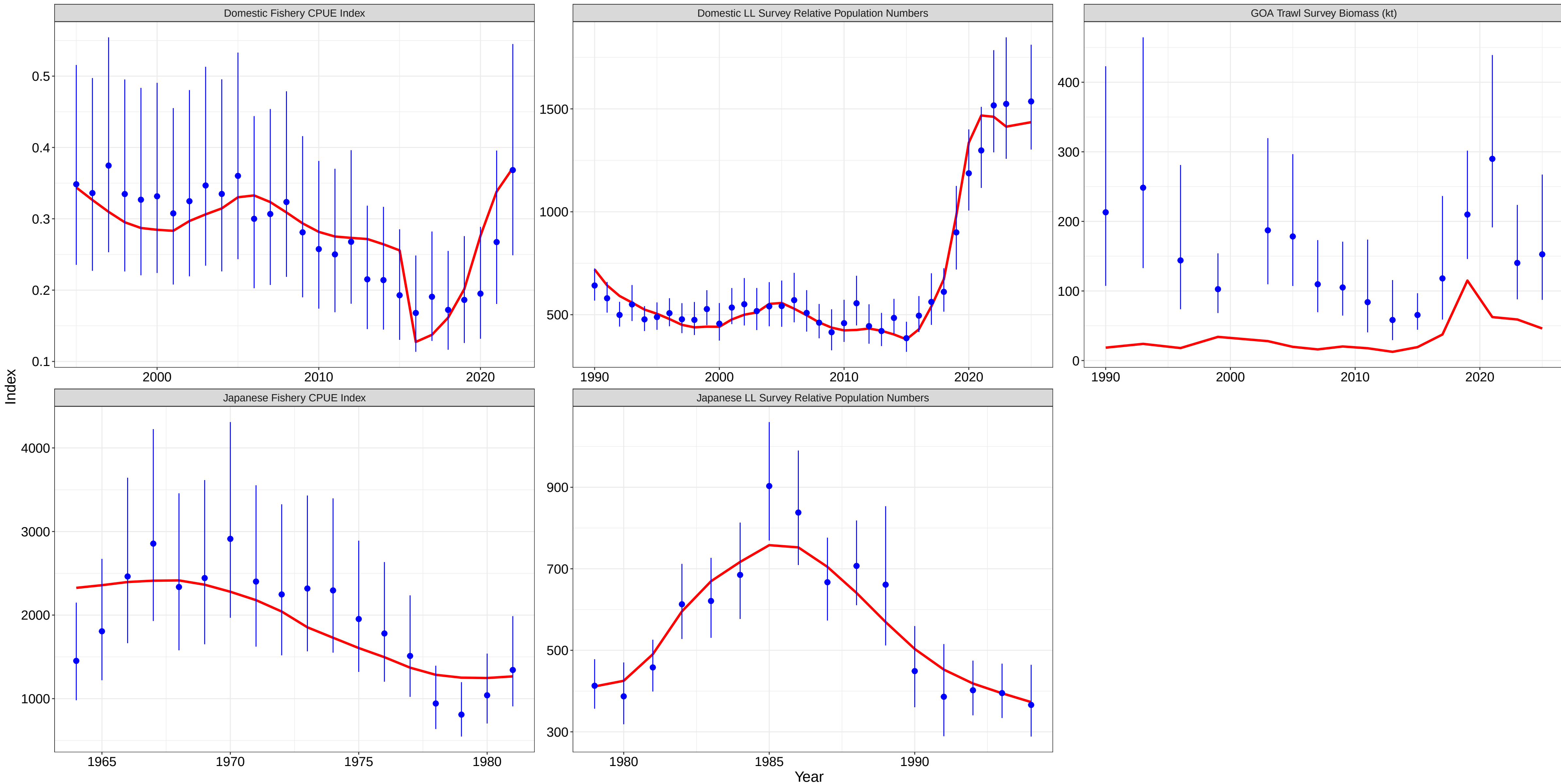
Model — 1 Year  20 40 60



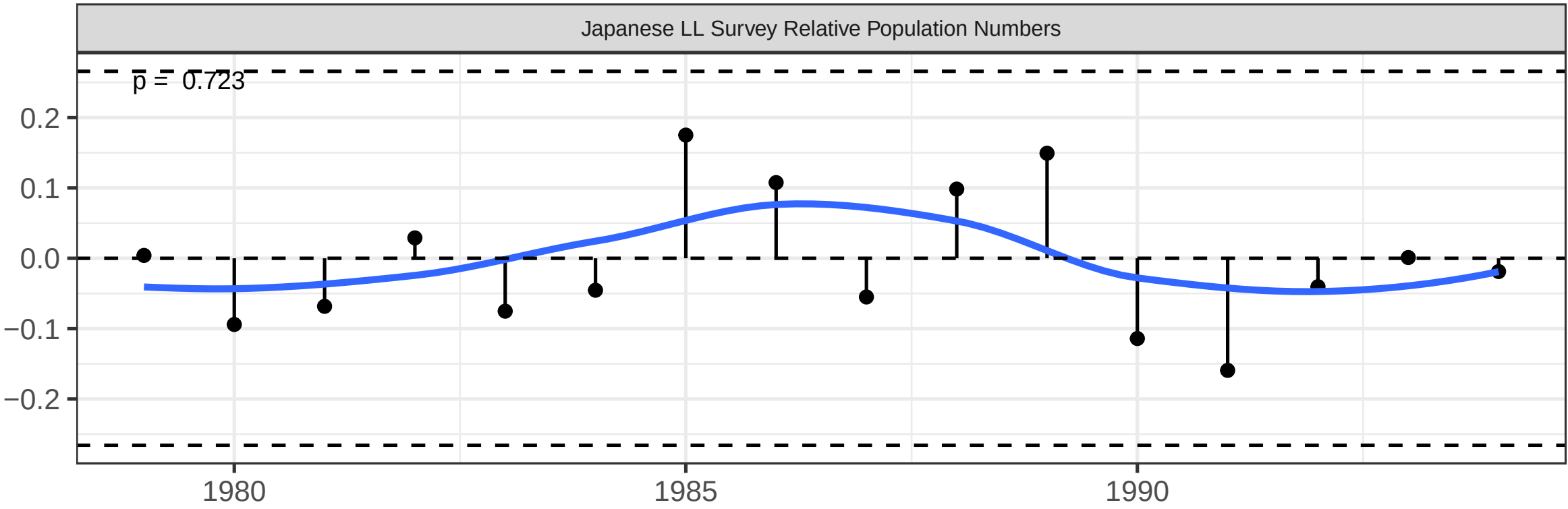
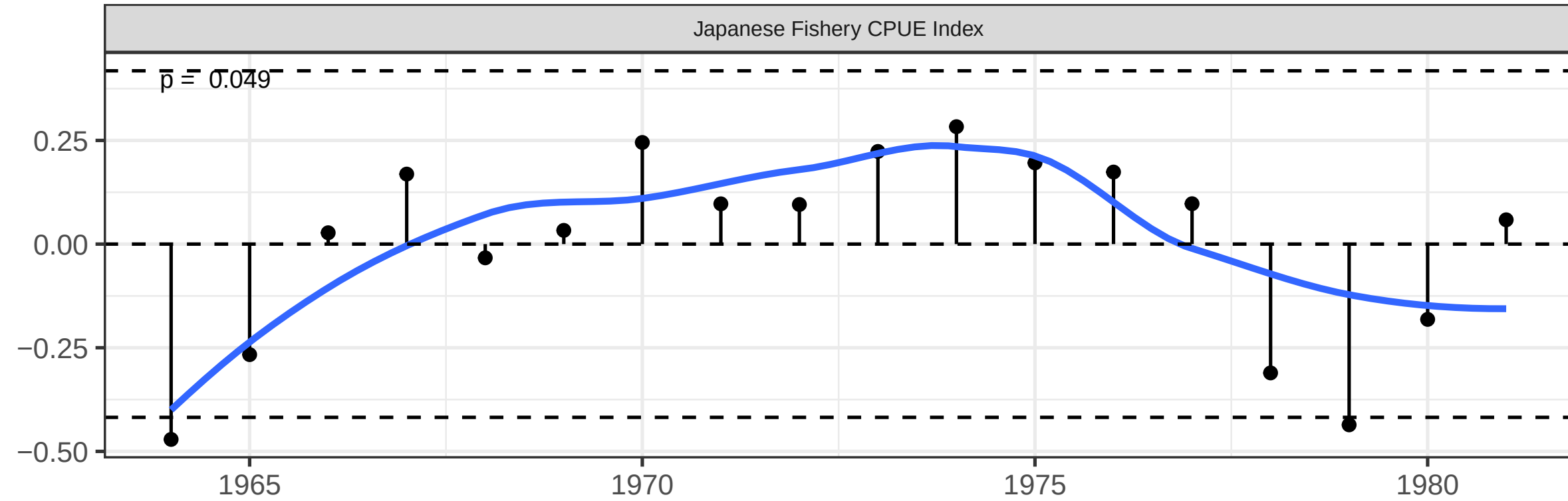
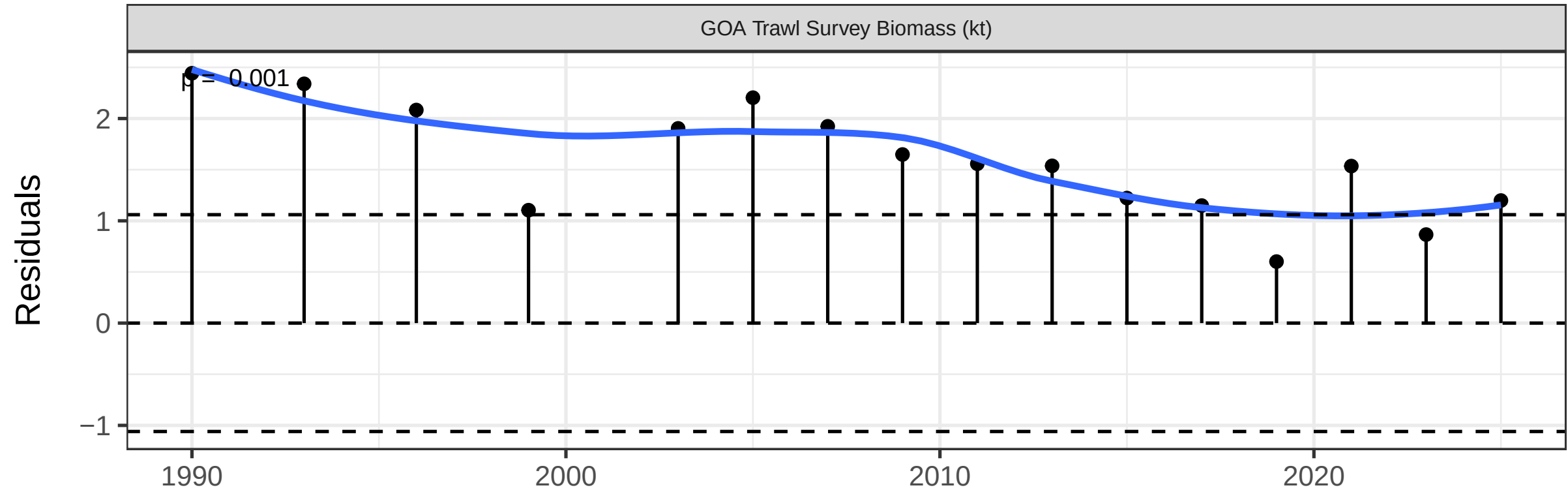
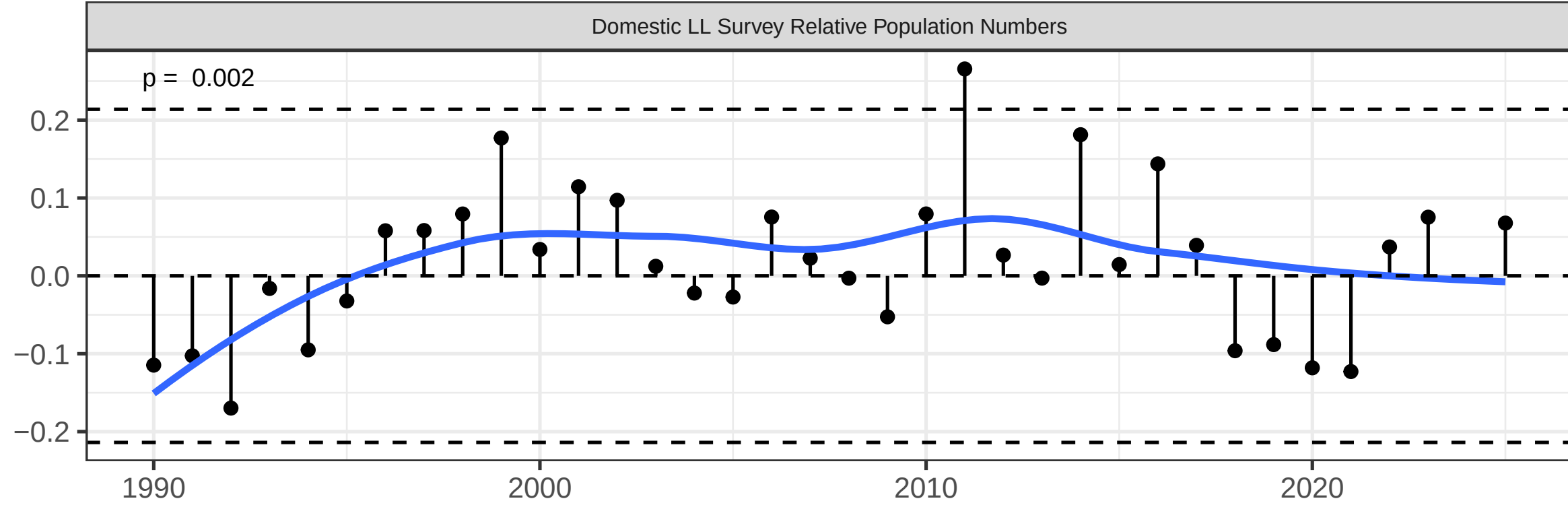
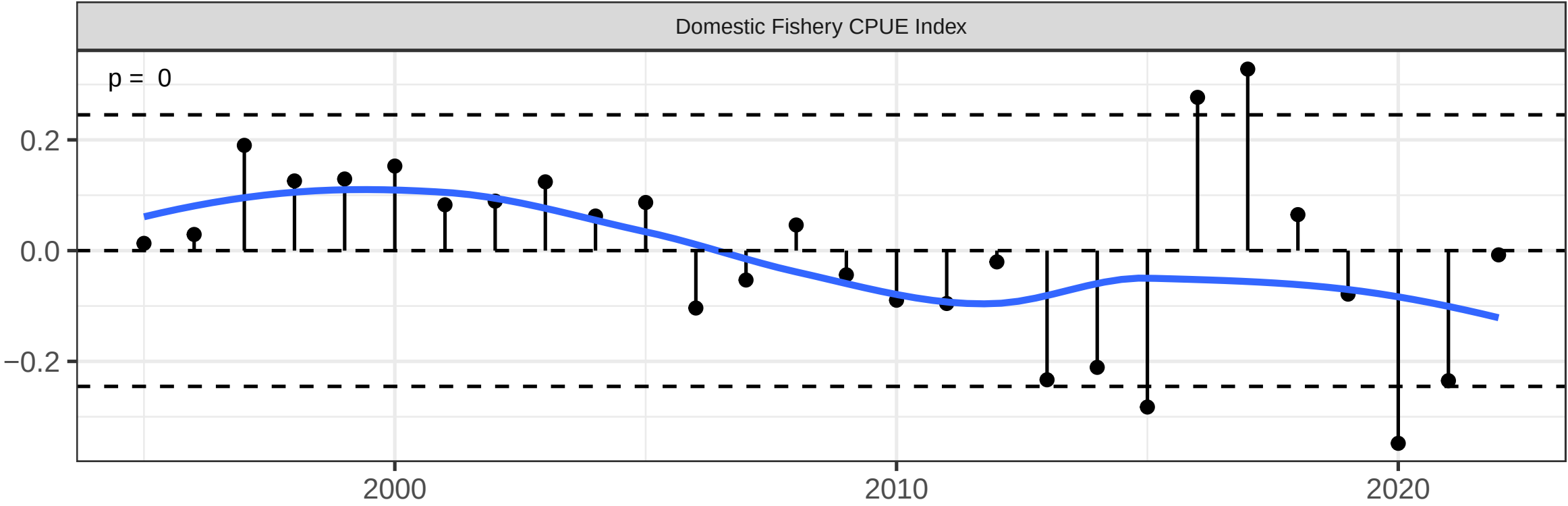




Index Fits

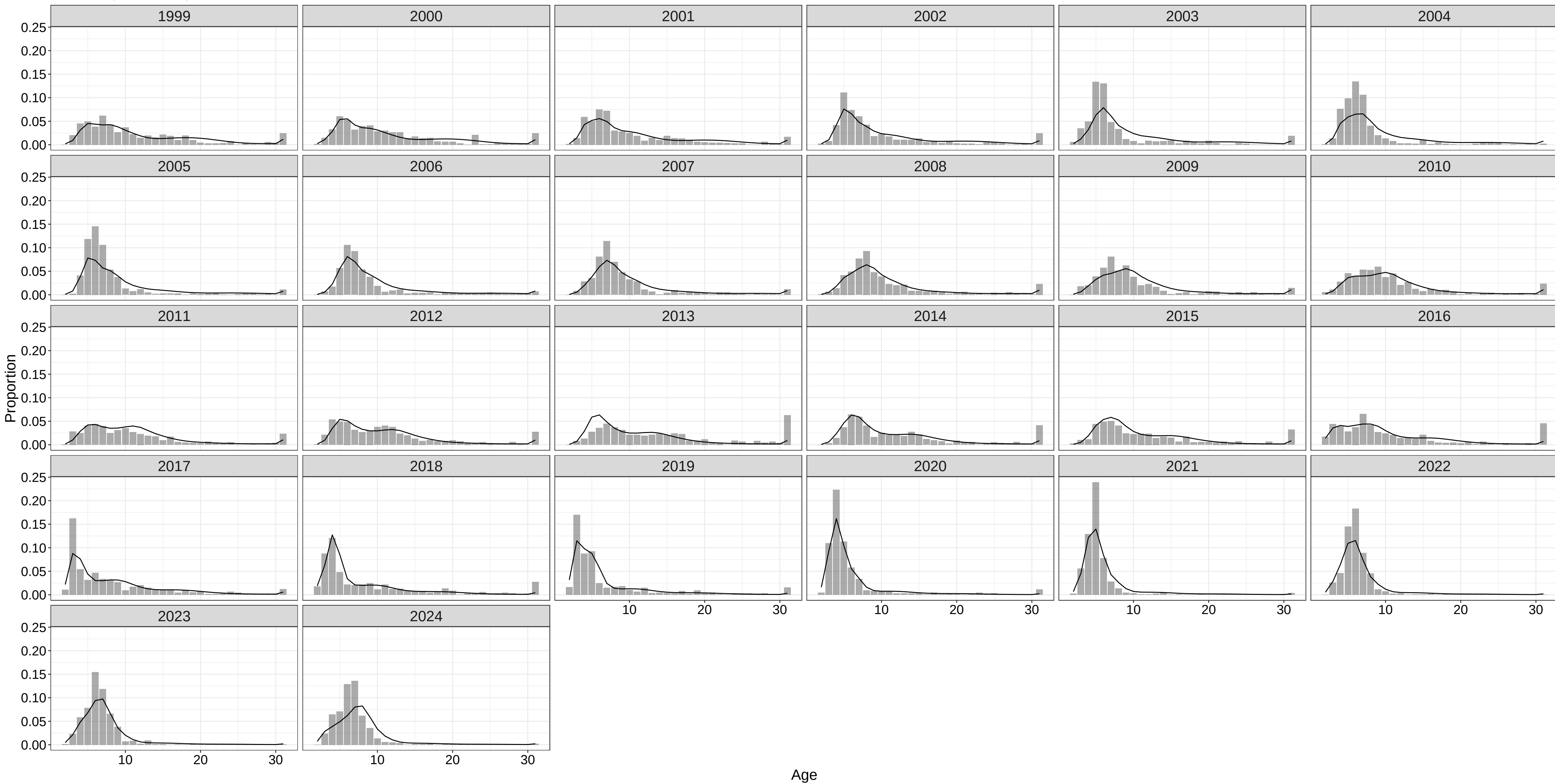


Index Residuals Runs Test

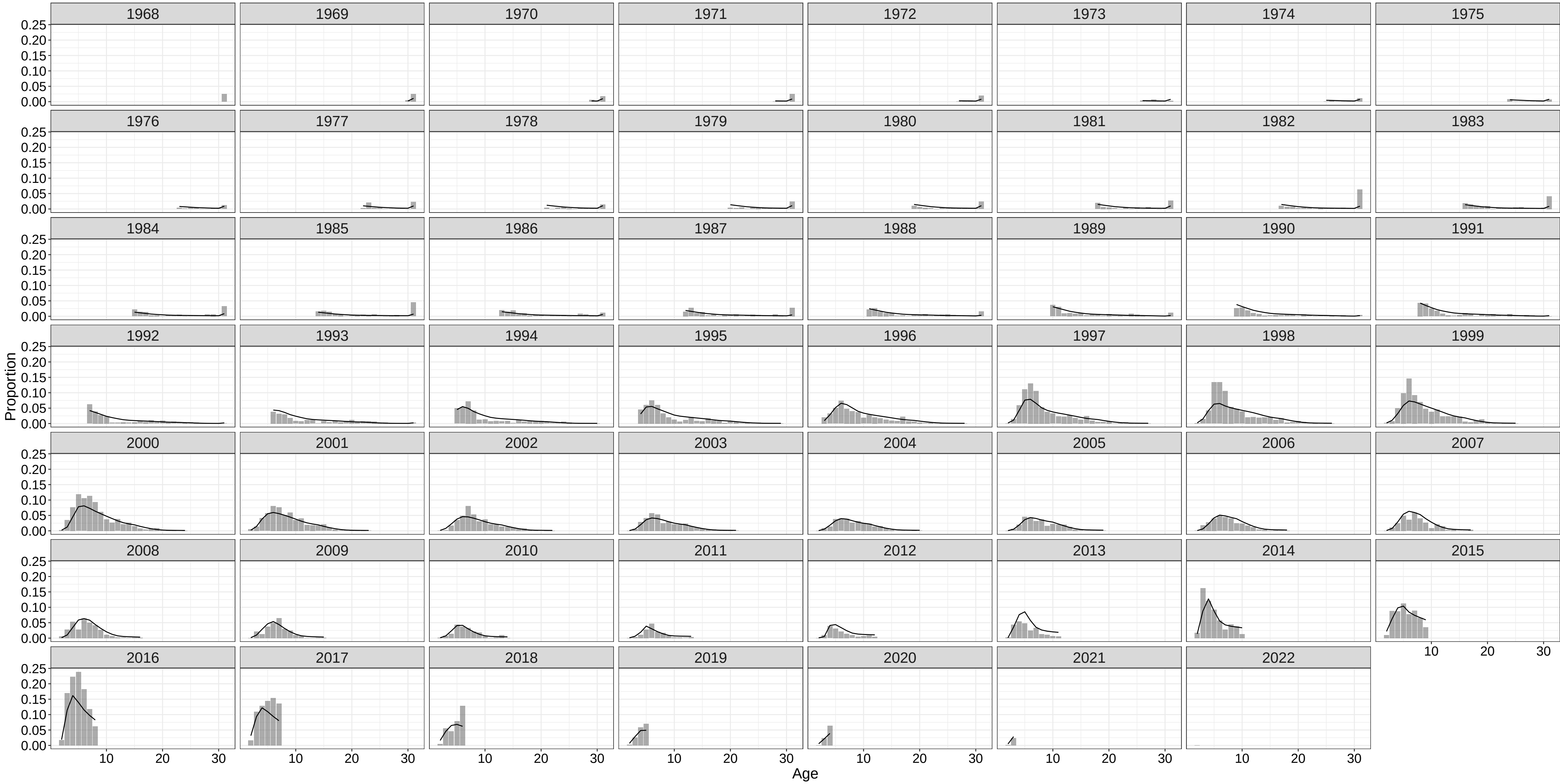


Year

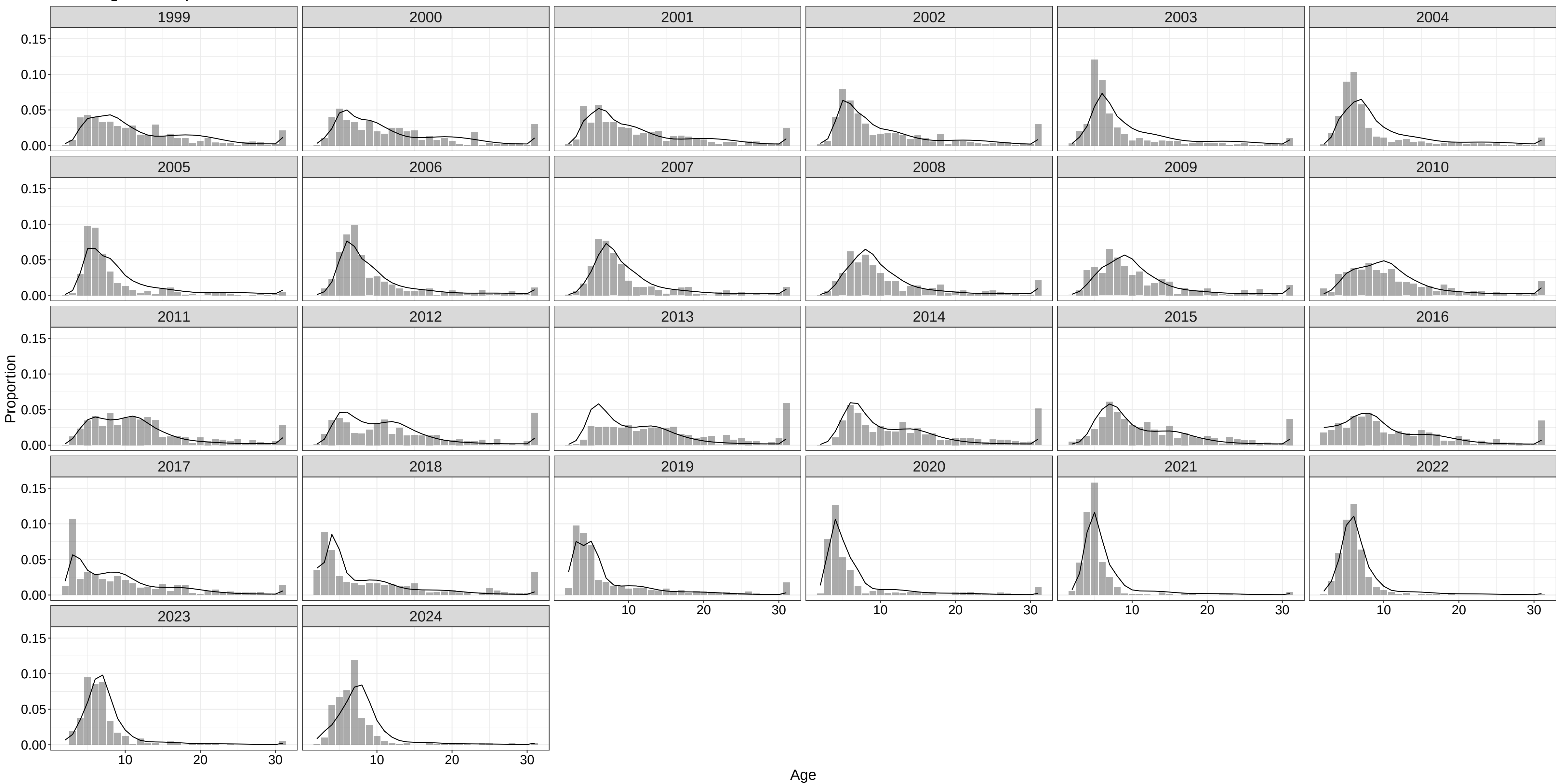
LLF Age Comps Female



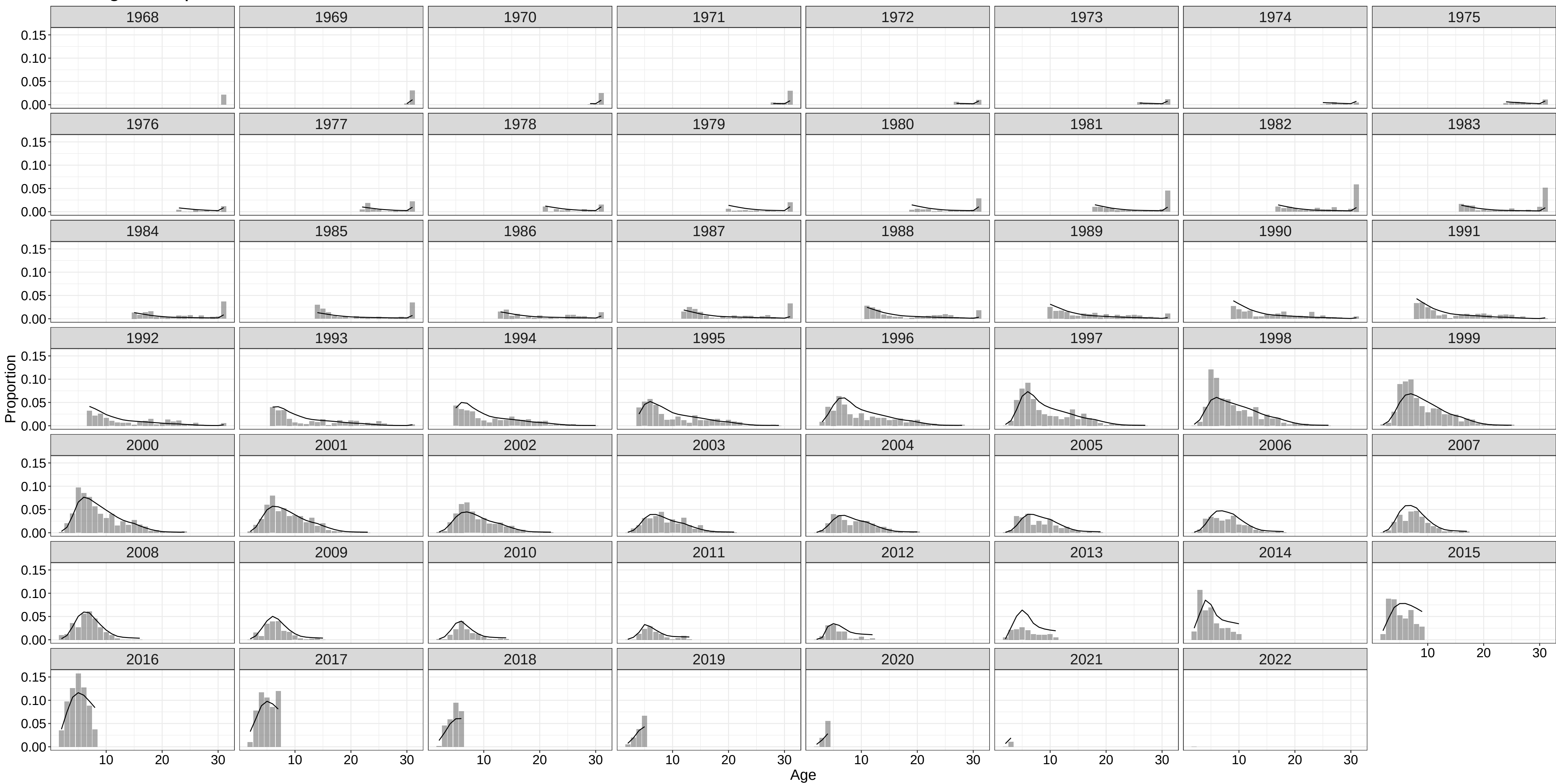
LLF Age Comps Cohort Fits Female



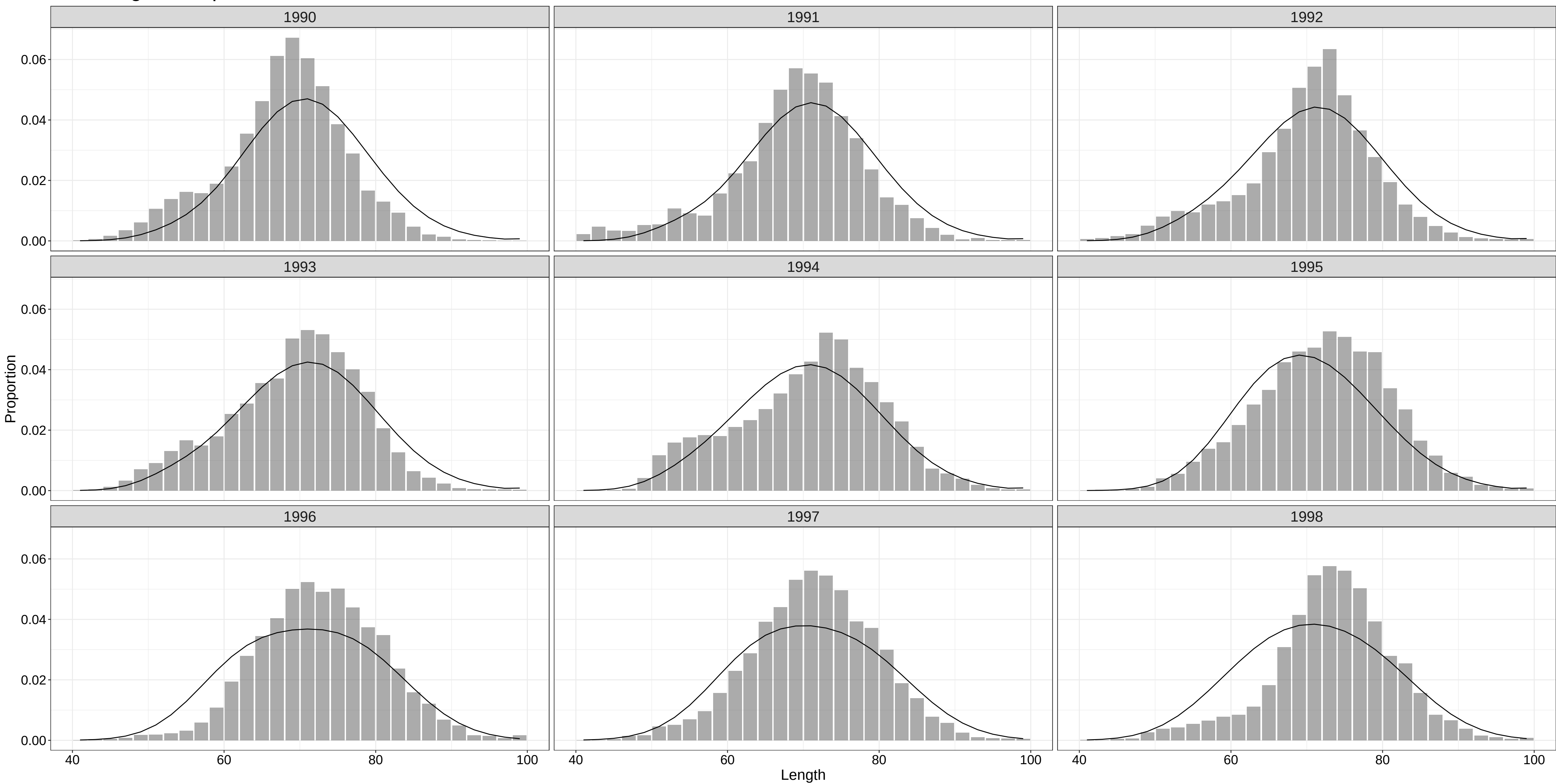
LLF Age Comps Male



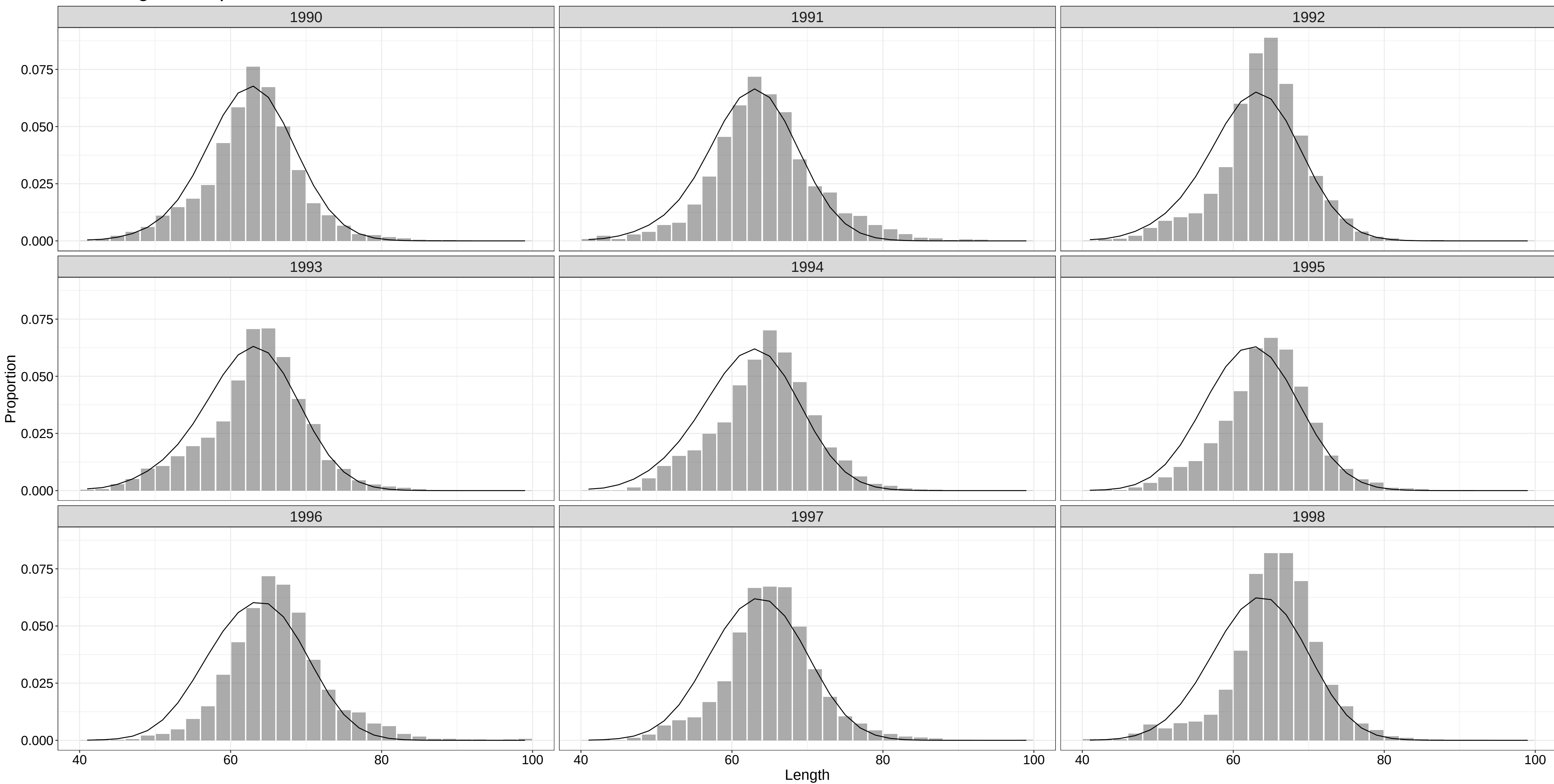
LLF Age Comps Cohort Fits Male



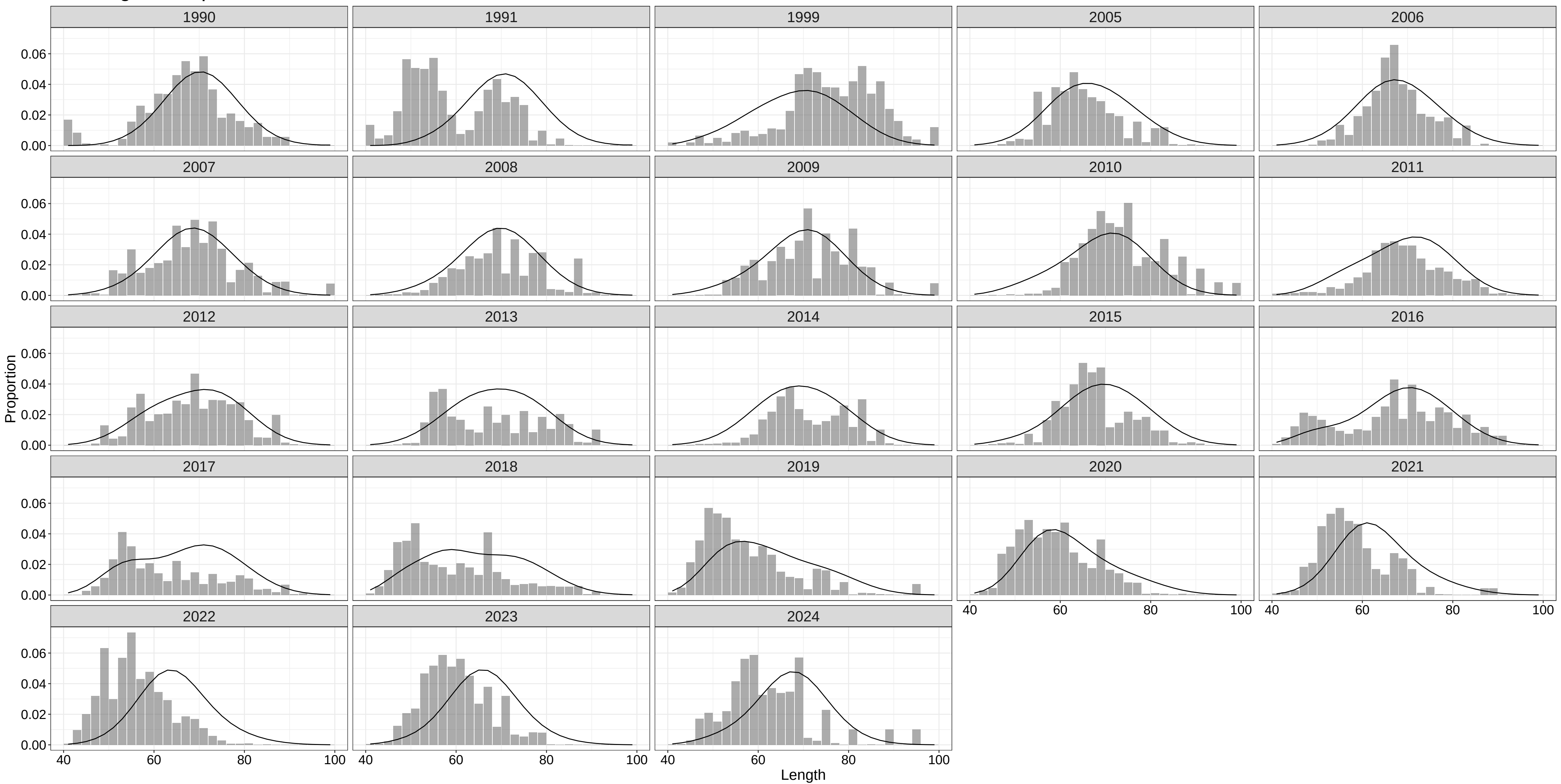
LLF Length Comps Female



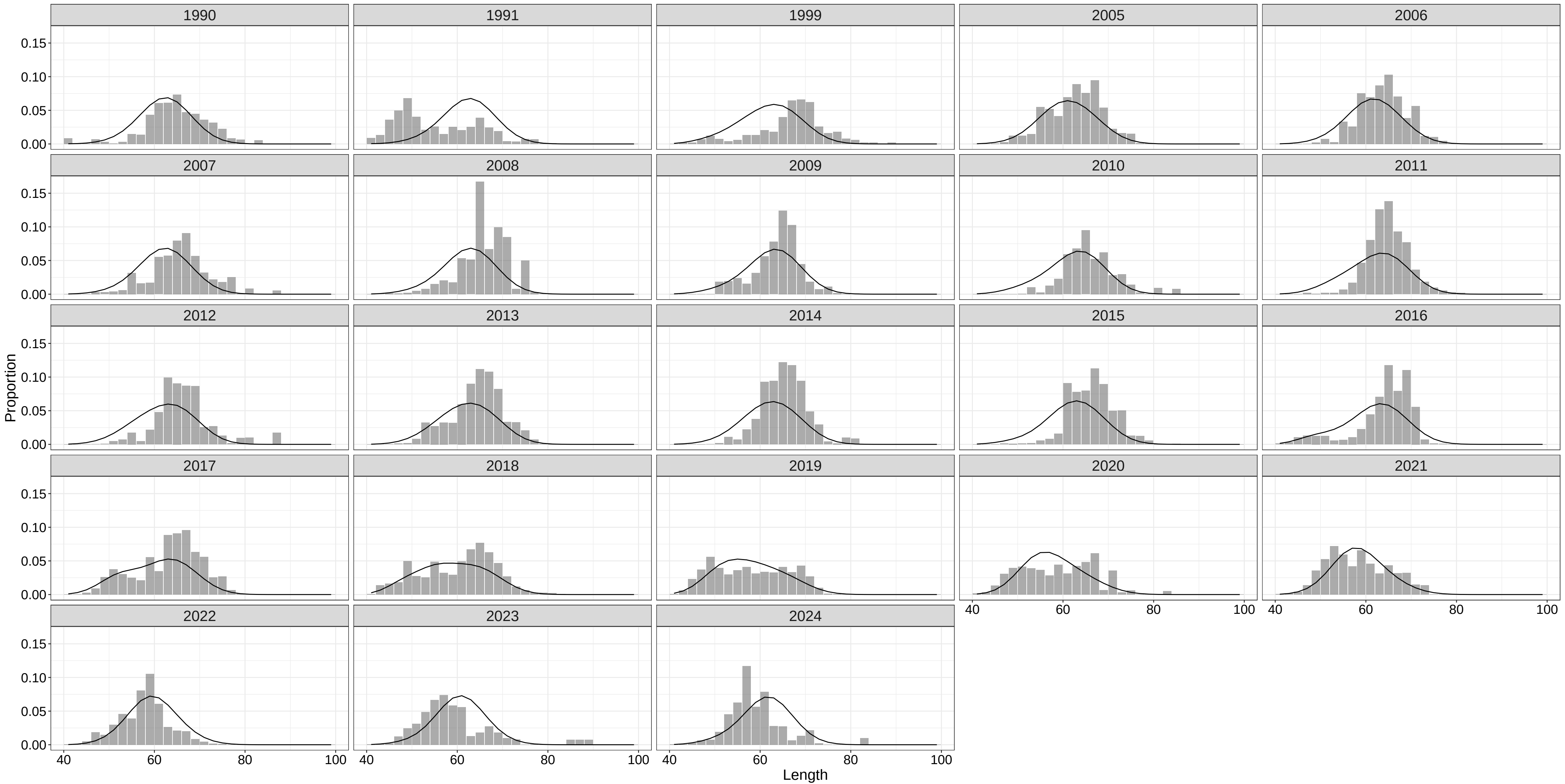
LLF Length Comps Male



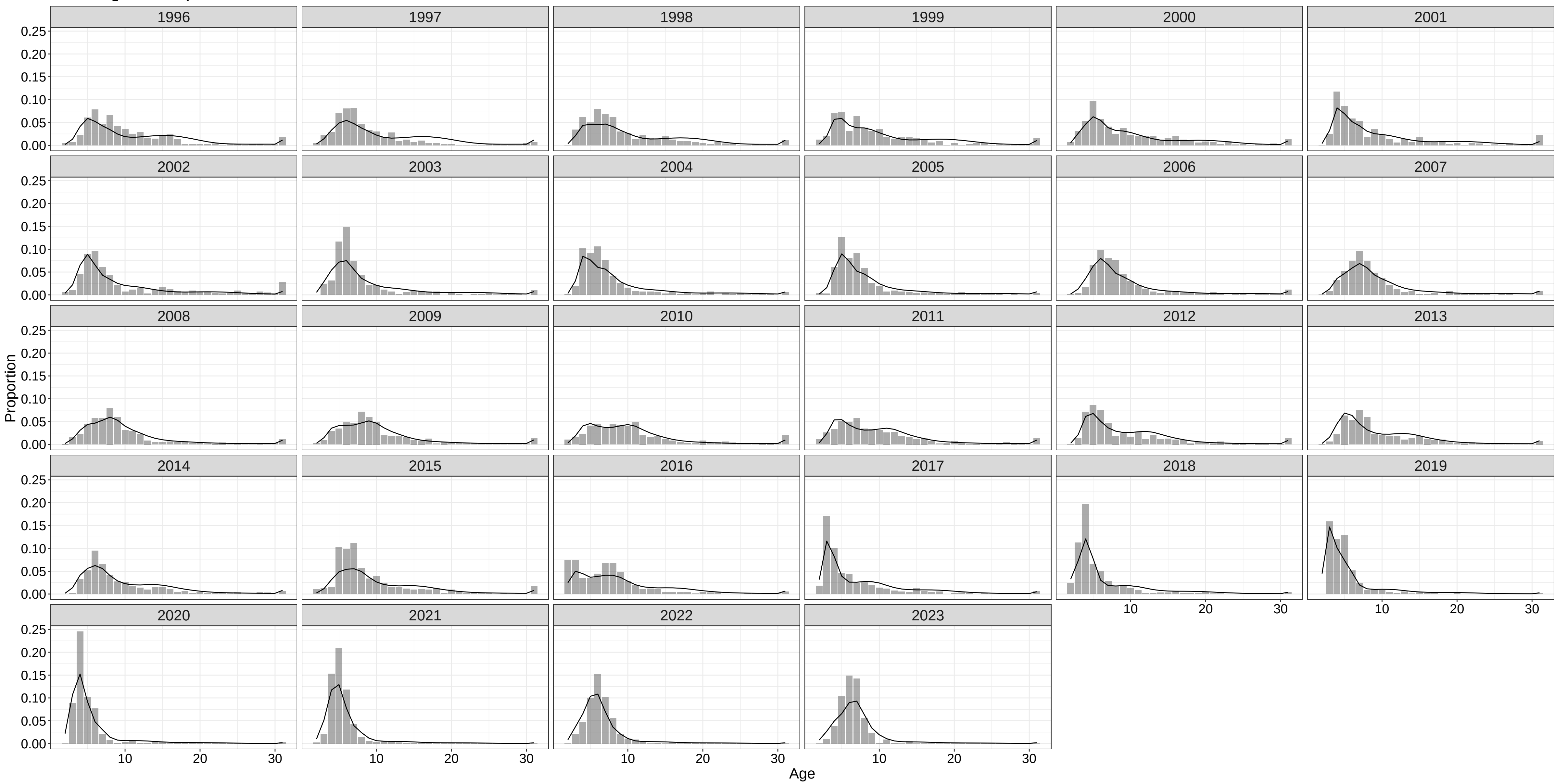
TF Length Comps Female



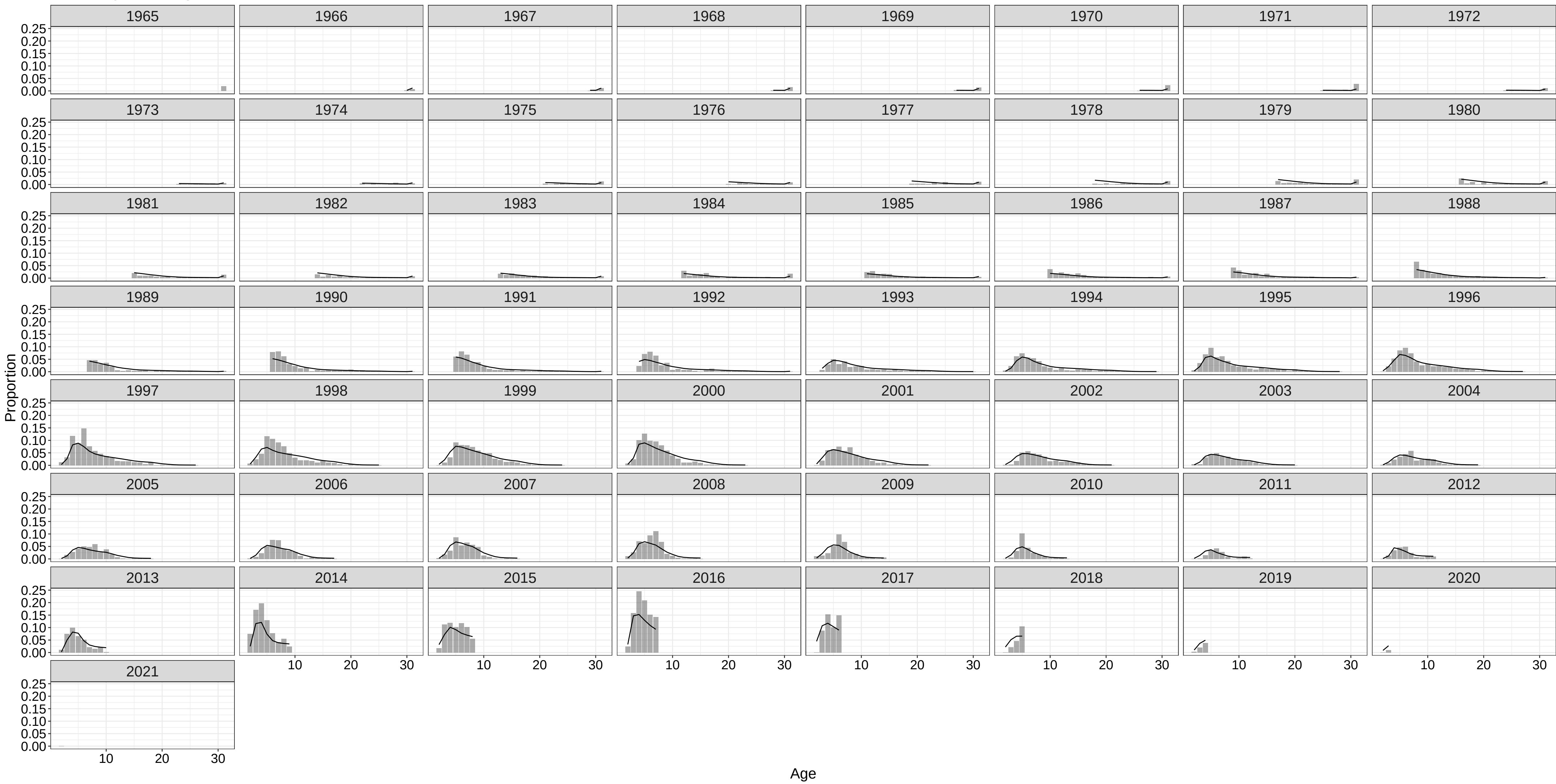
TF Length Comps Male



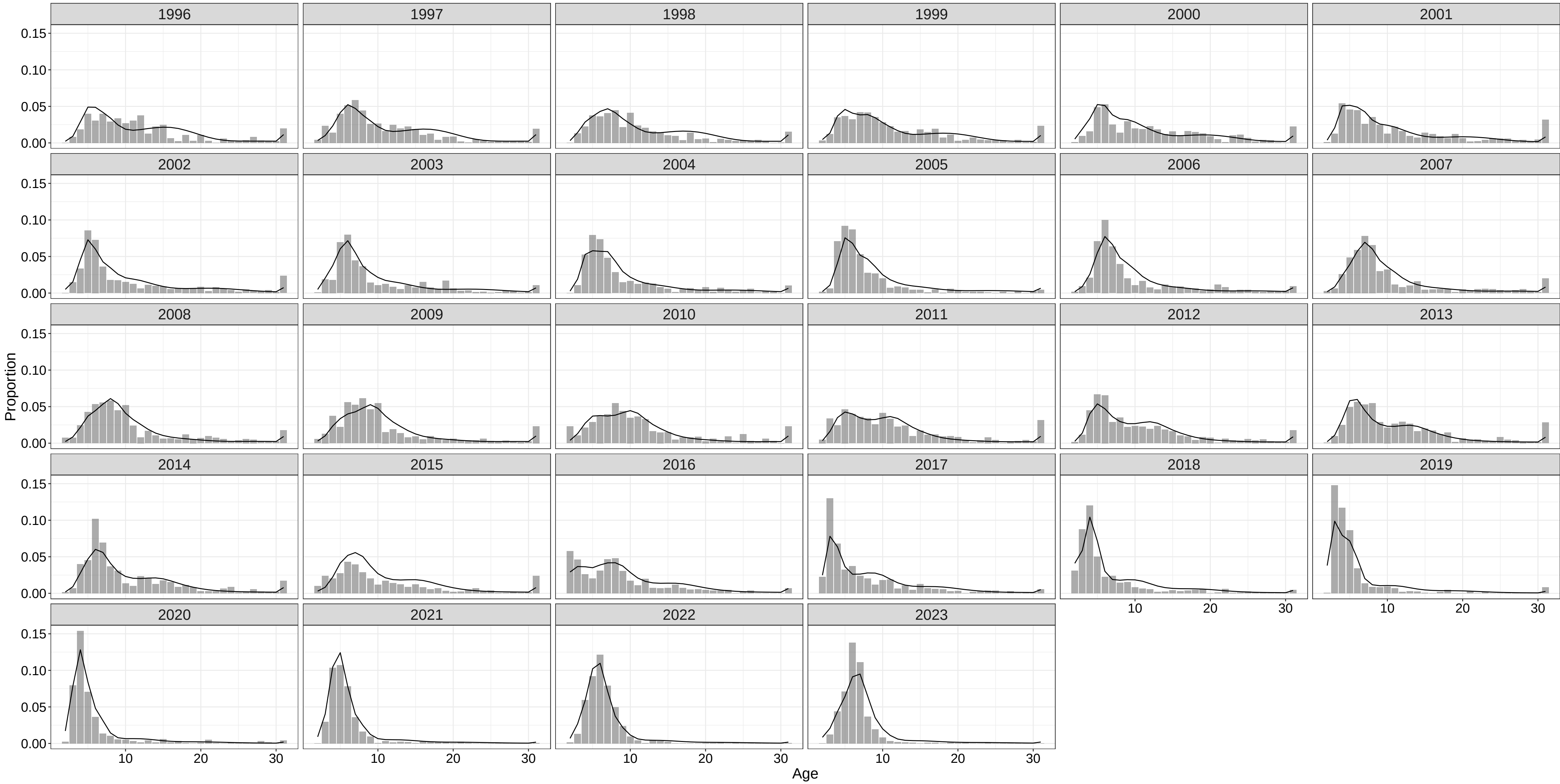
LLS Age Comps Female



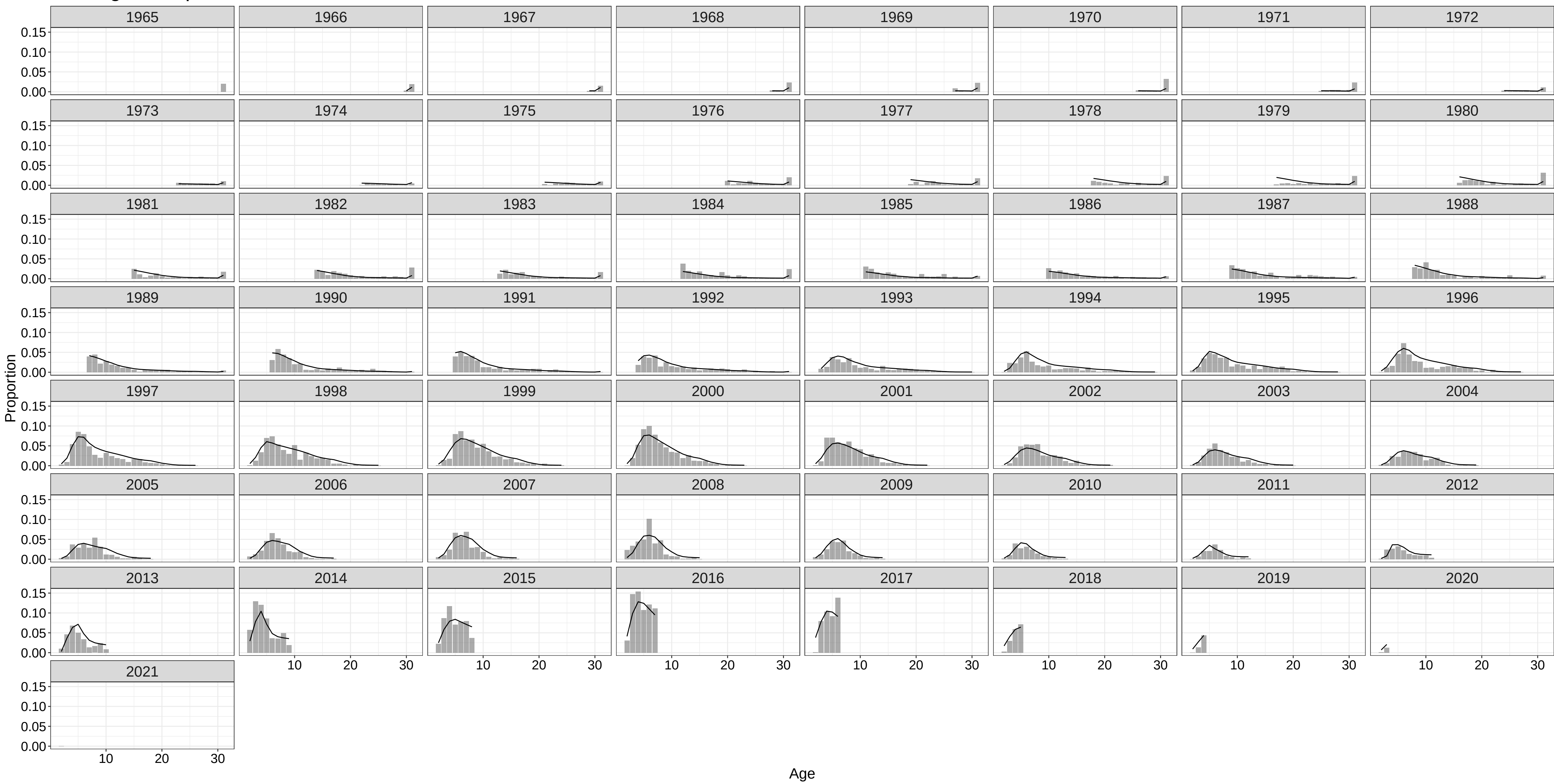
LLS Age Comps Cohort Fit Female



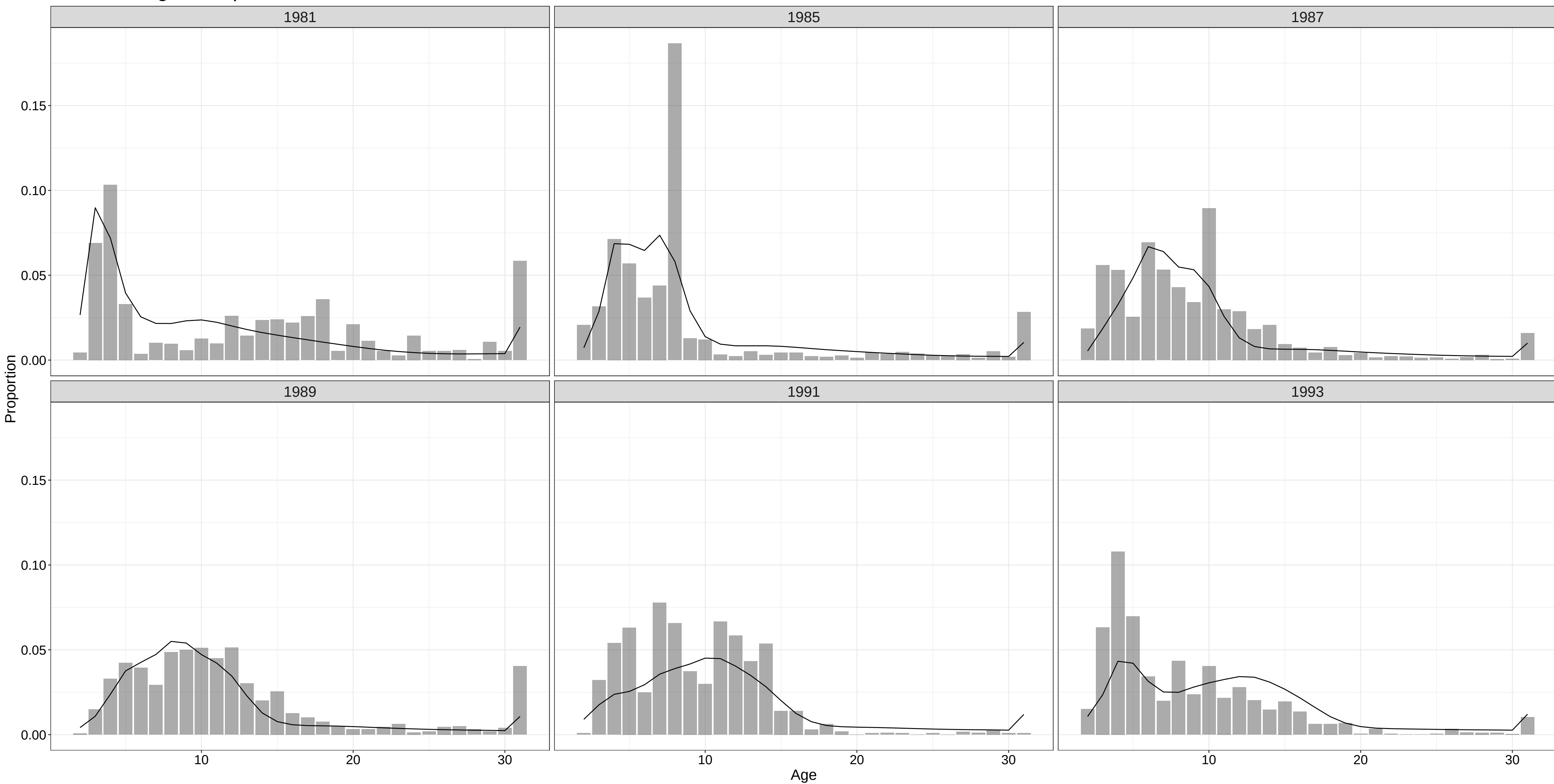
LLS Age Comps Male



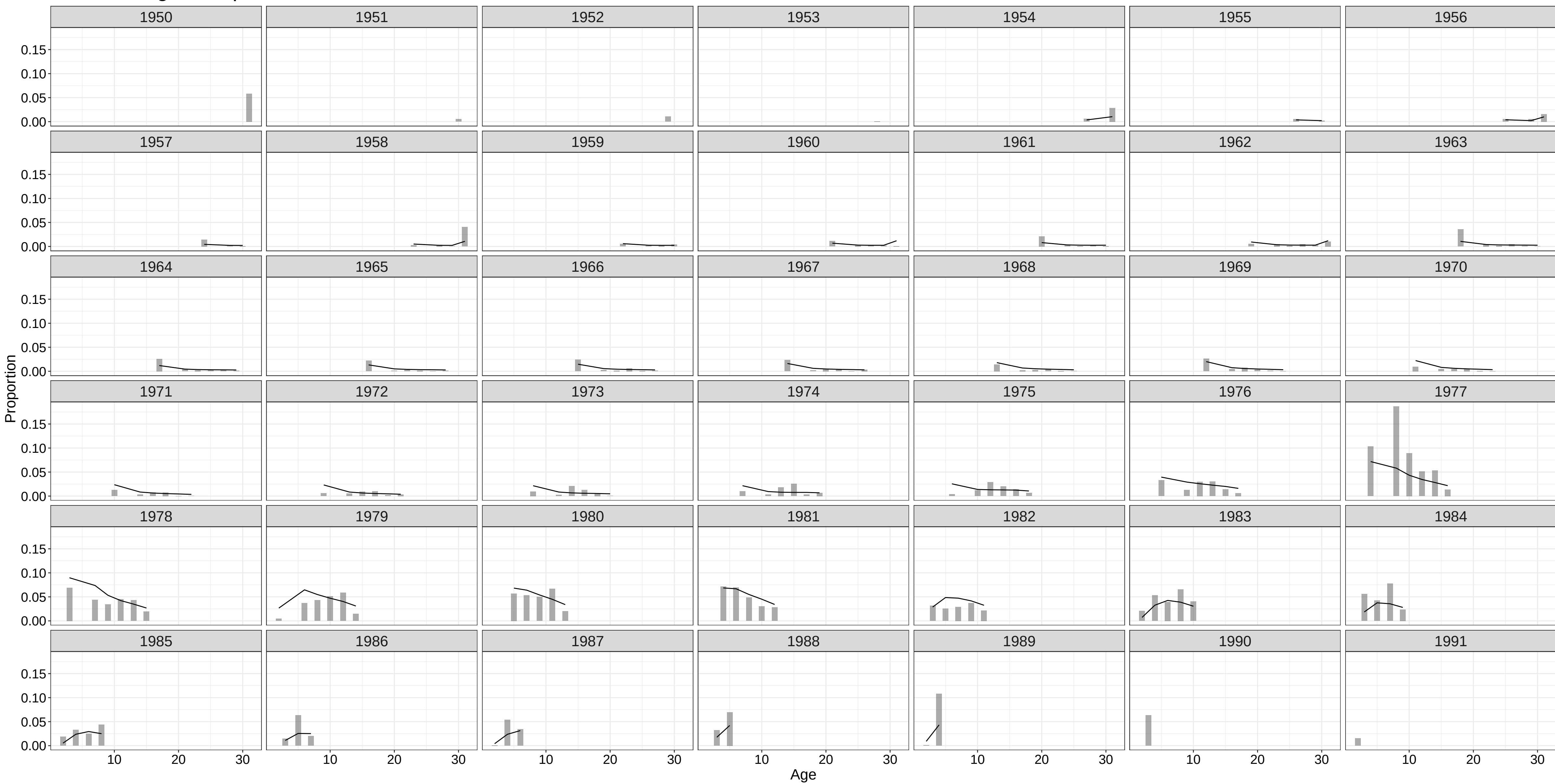
LLS Age Comps Cohort Fit Male



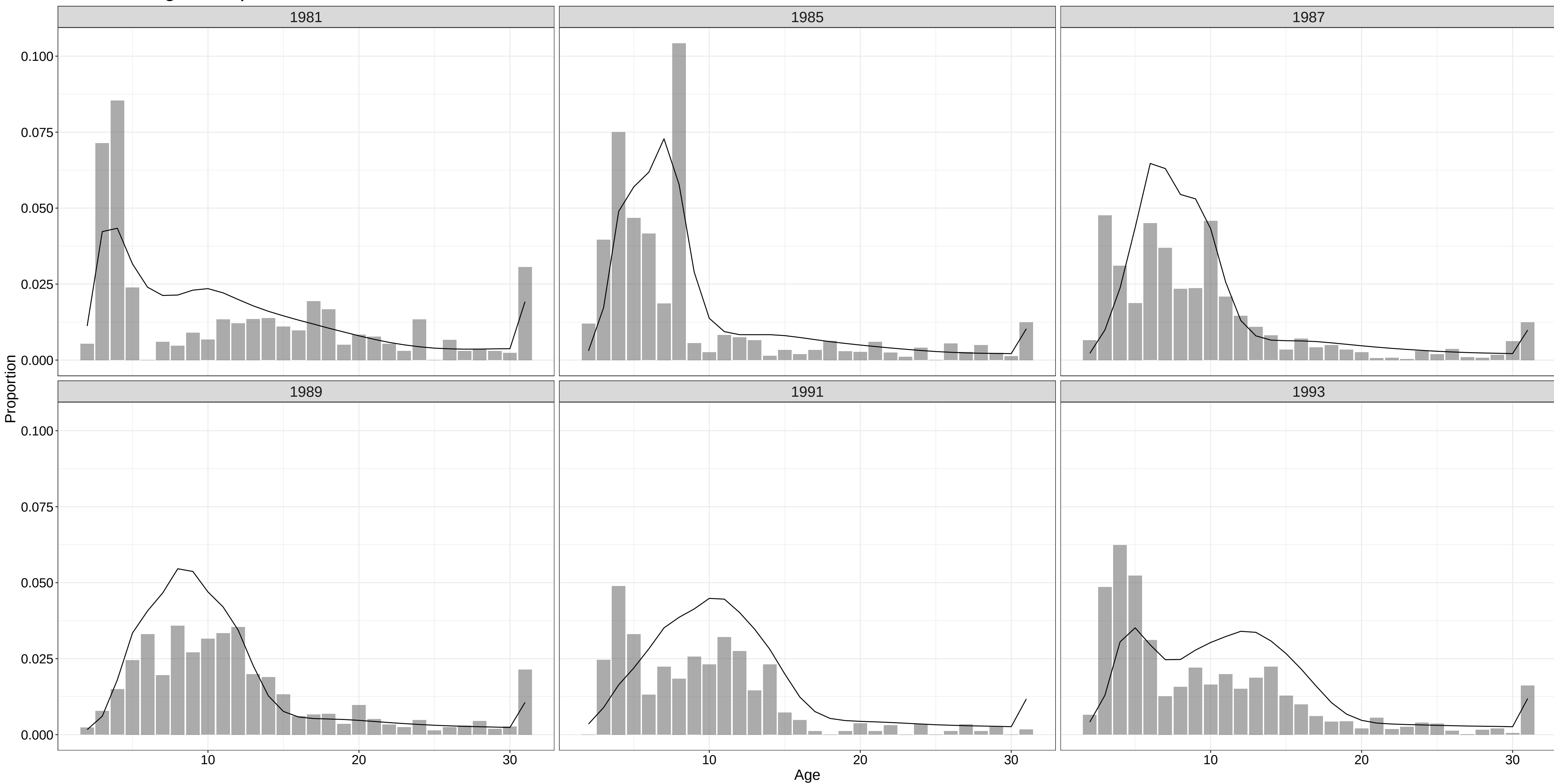
JPN LLS Age Comps Female



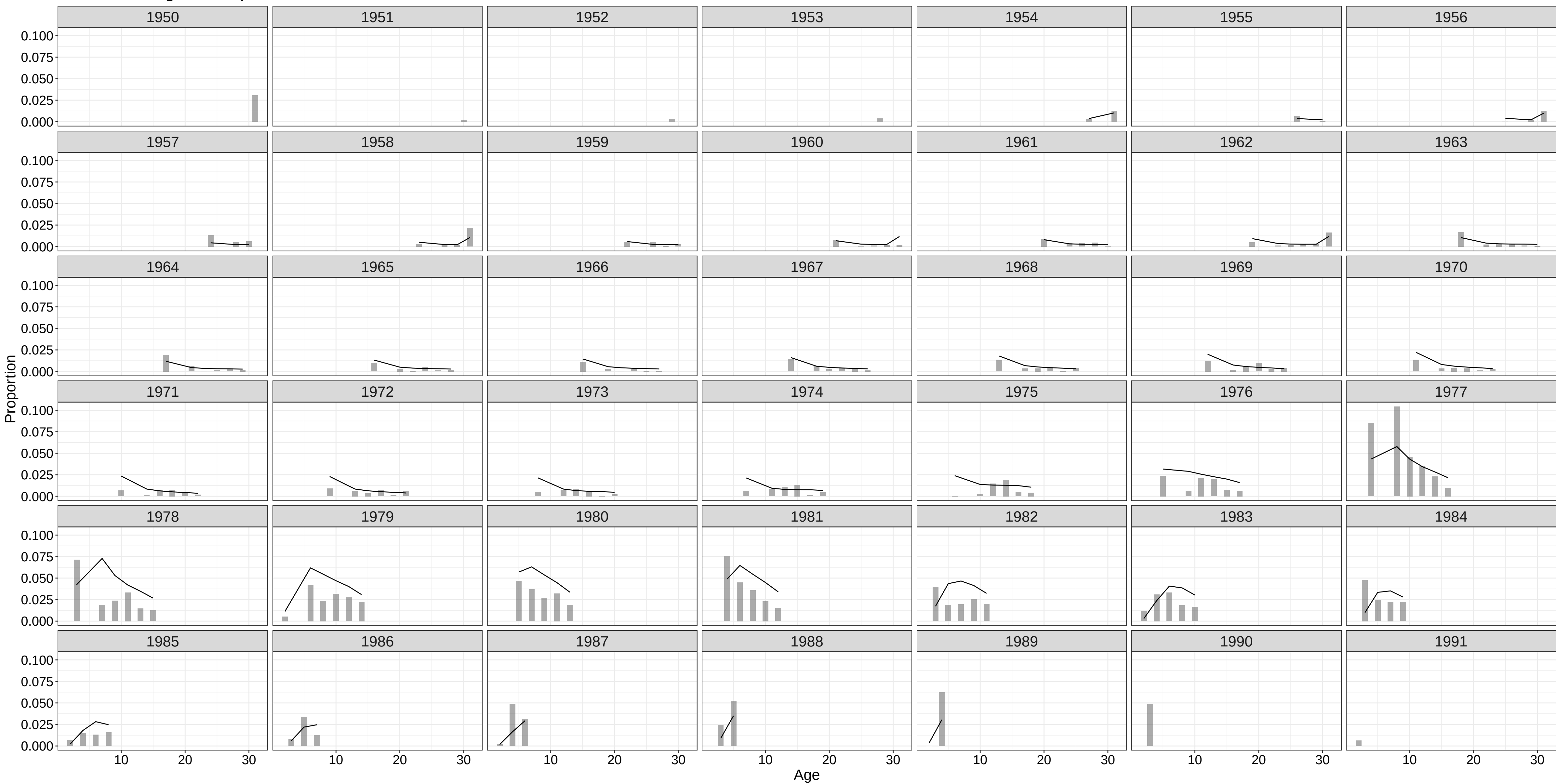
JPN LLS Age Comps Cohort Fit Female



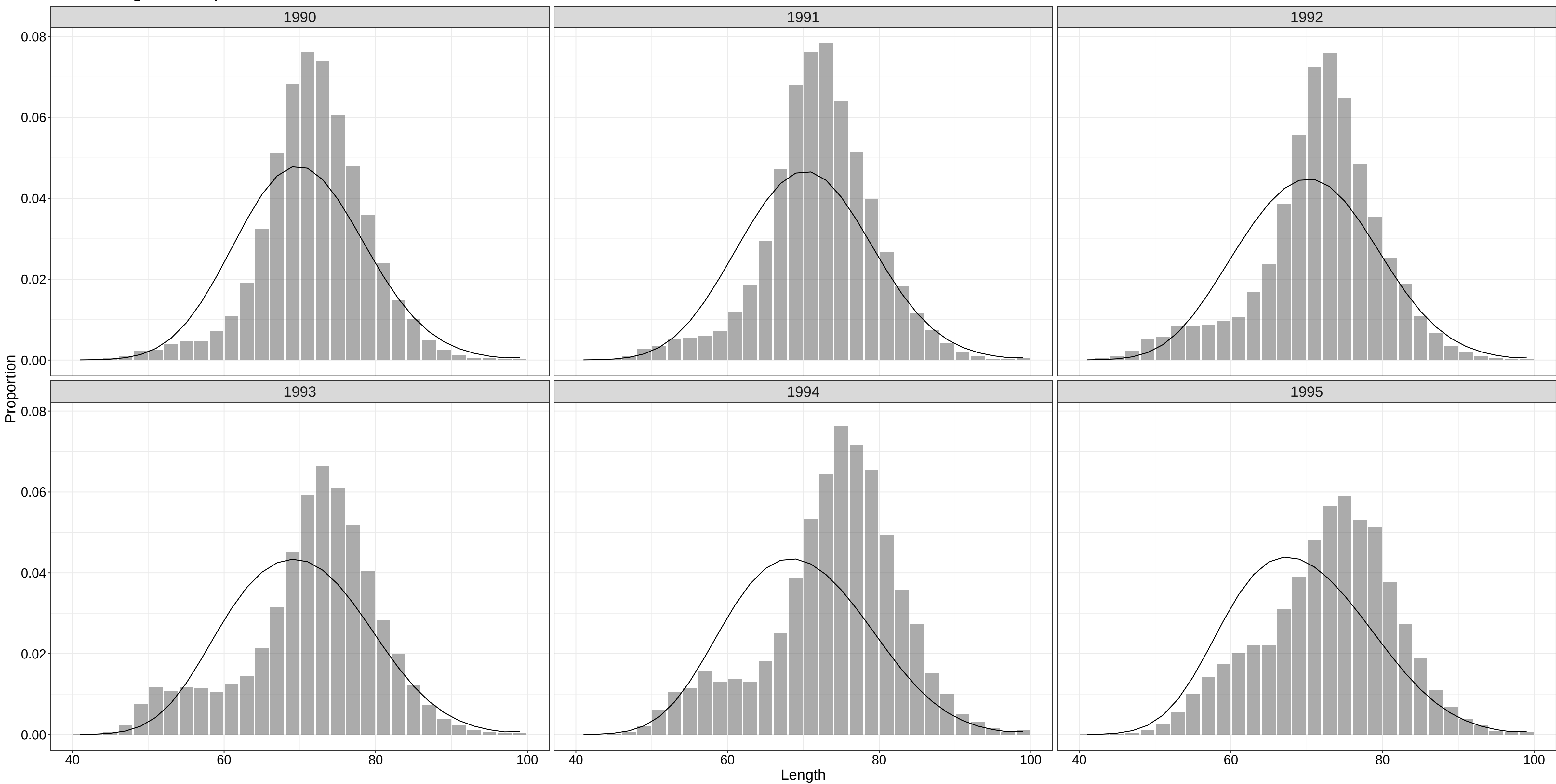
JPN LLS Age Comps Male



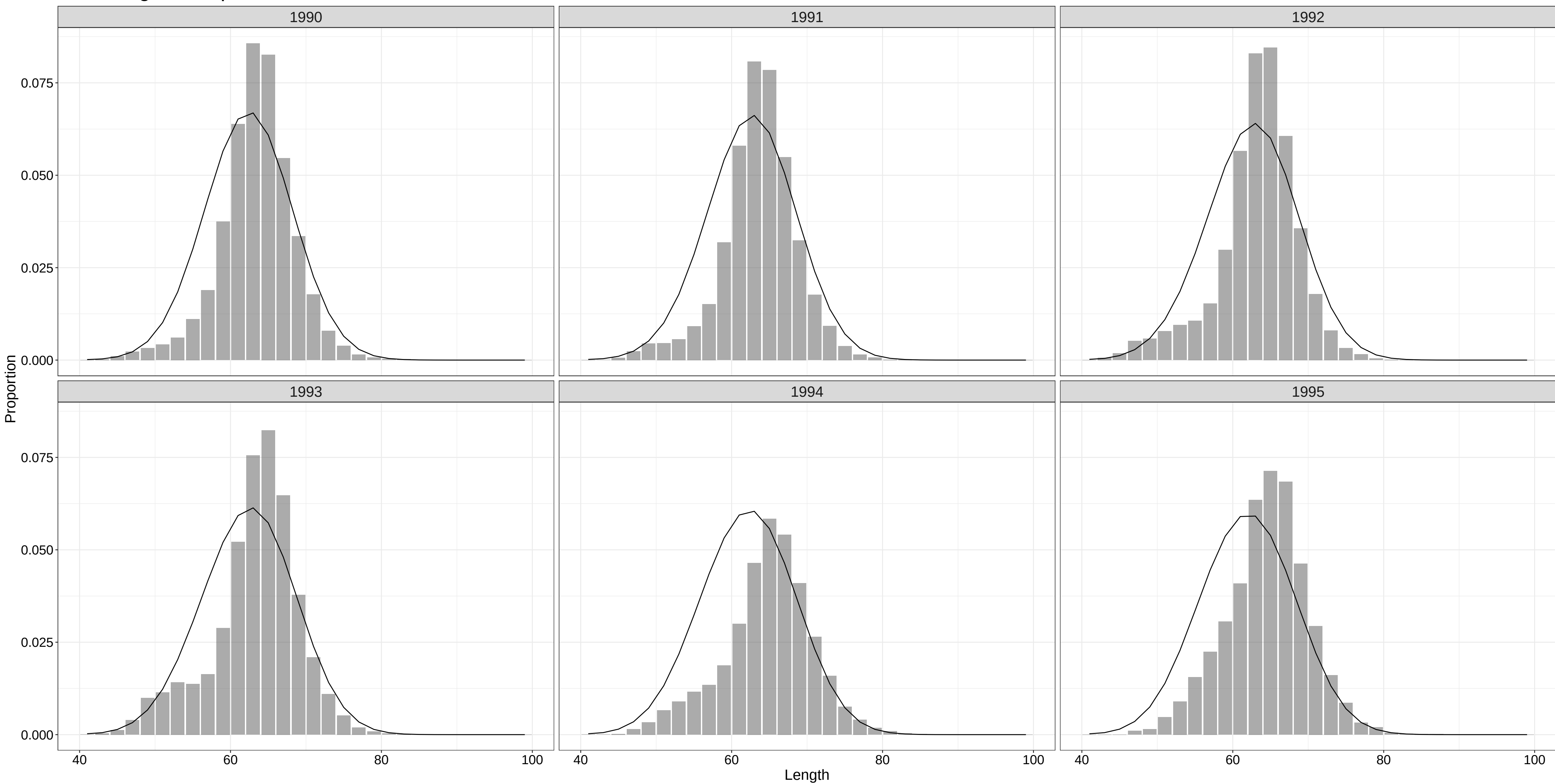
JPN LLS Age Comps Cohort Fit Male



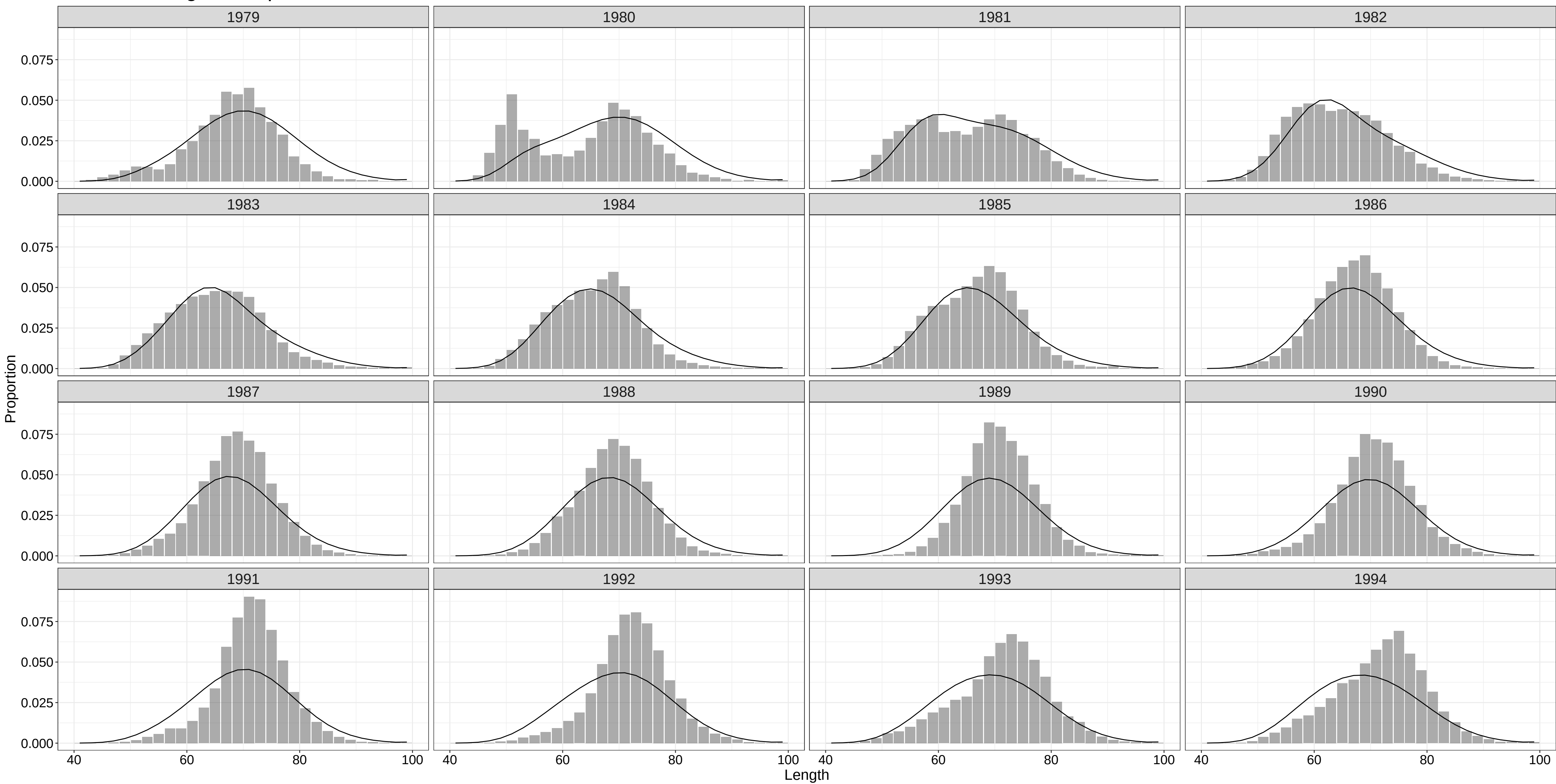
LLS Length Comps Female



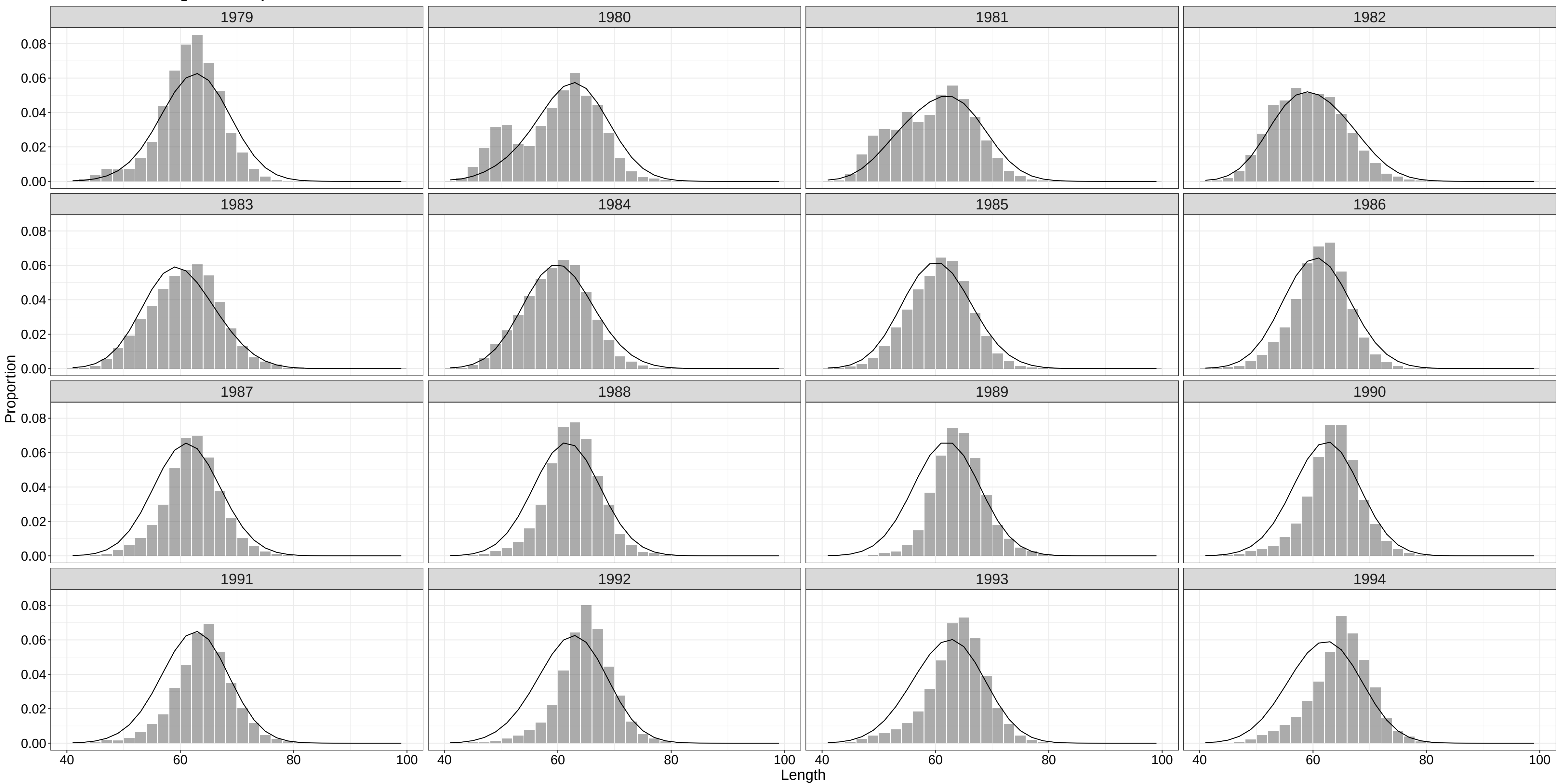
LLS Length Comps Male



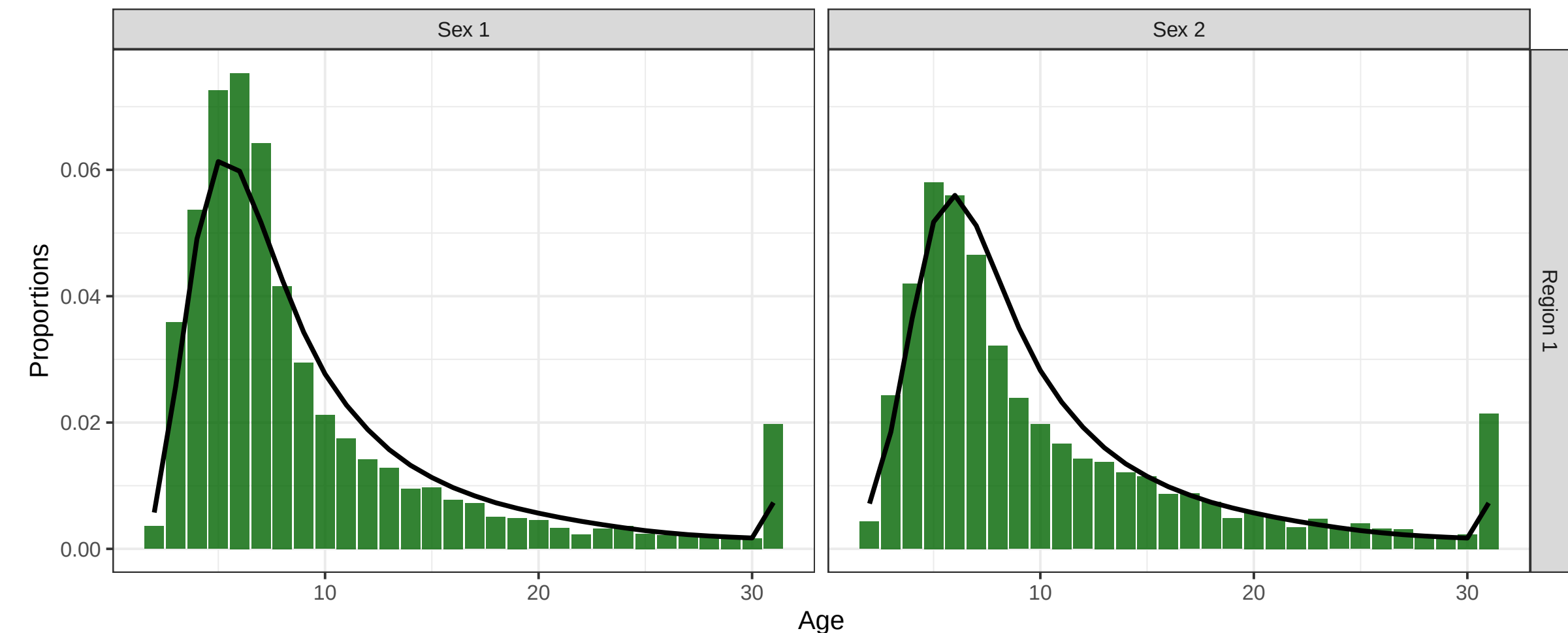
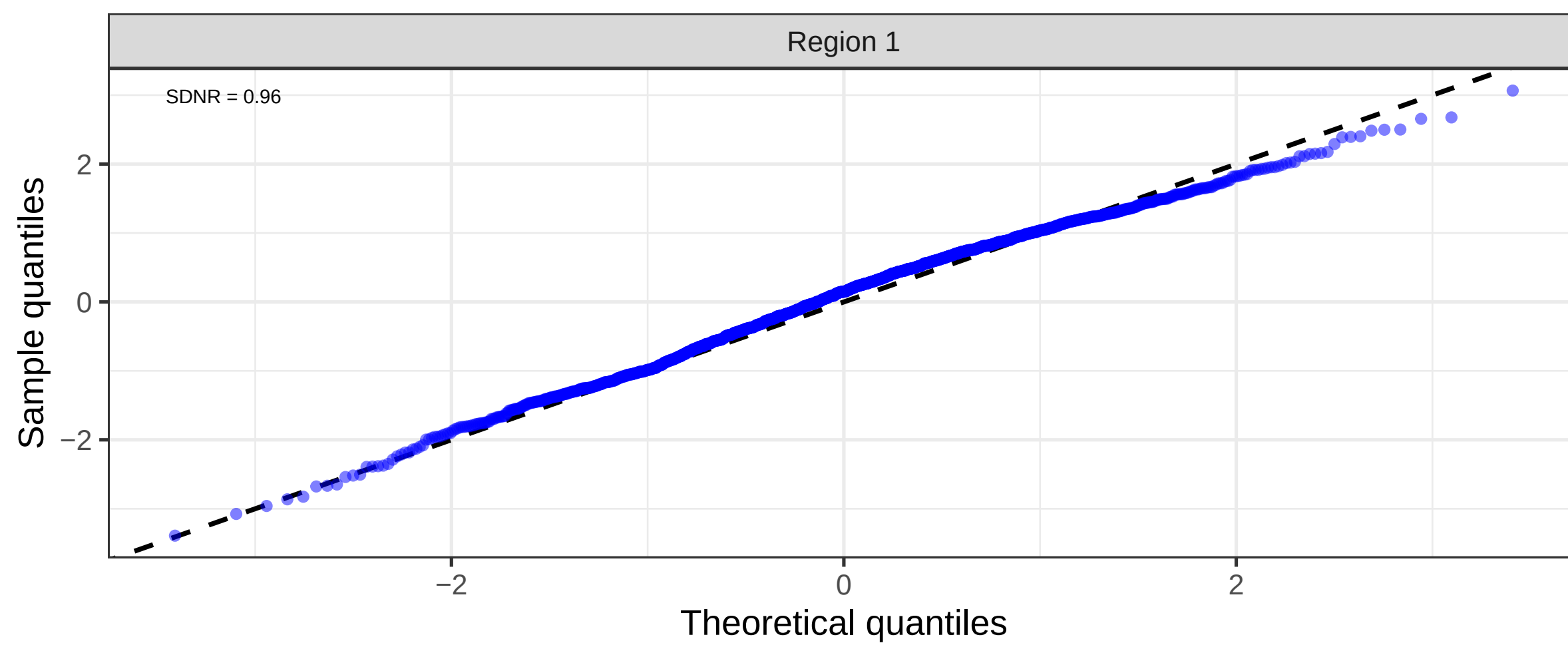
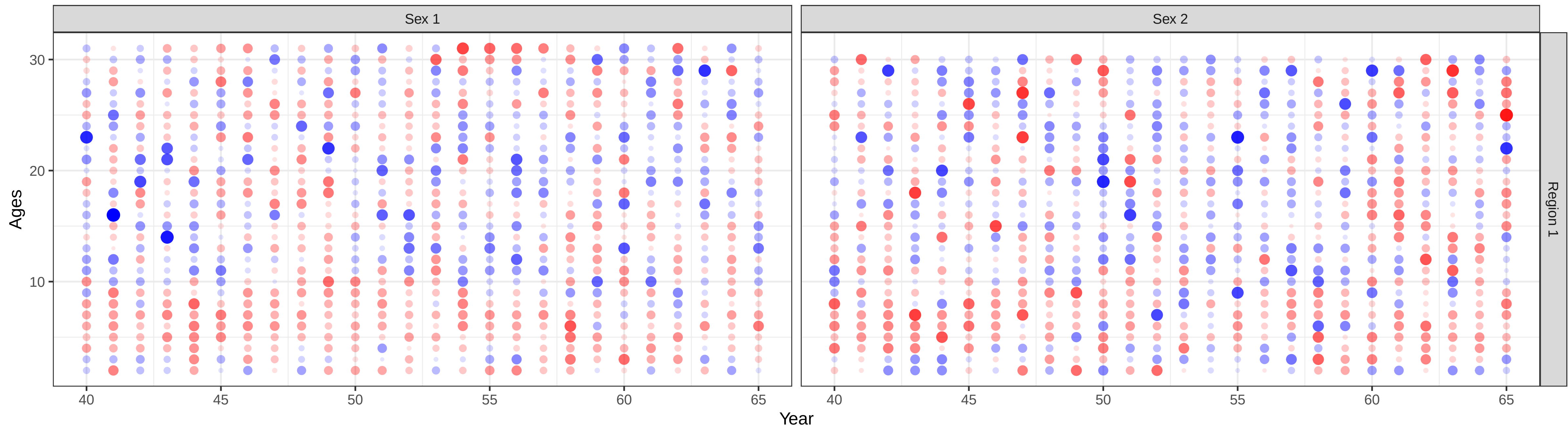
JPN LLS Length Comps Female

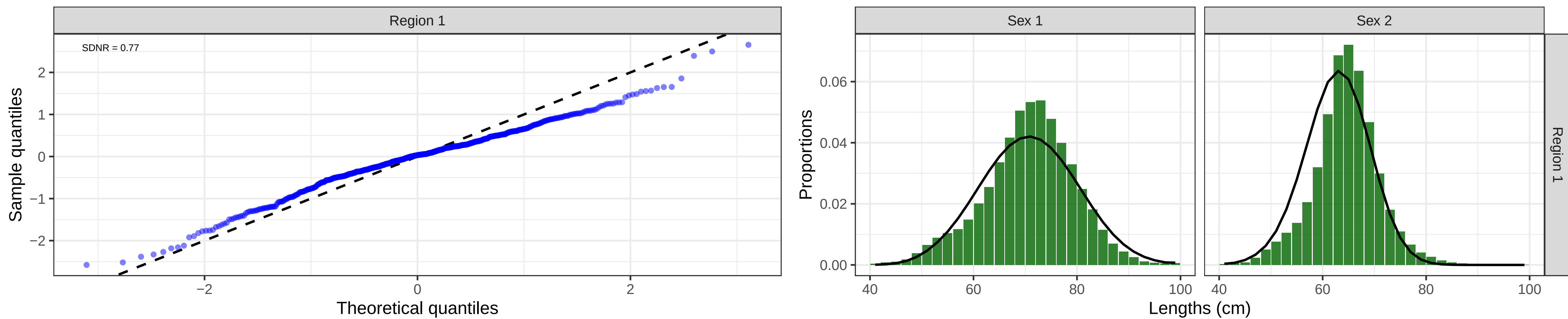
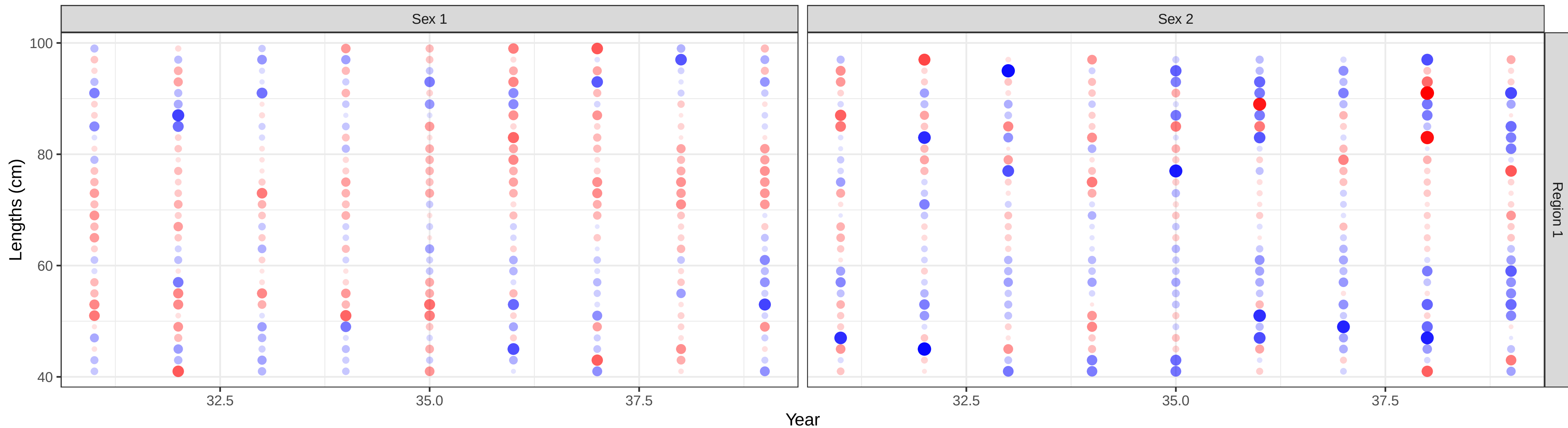


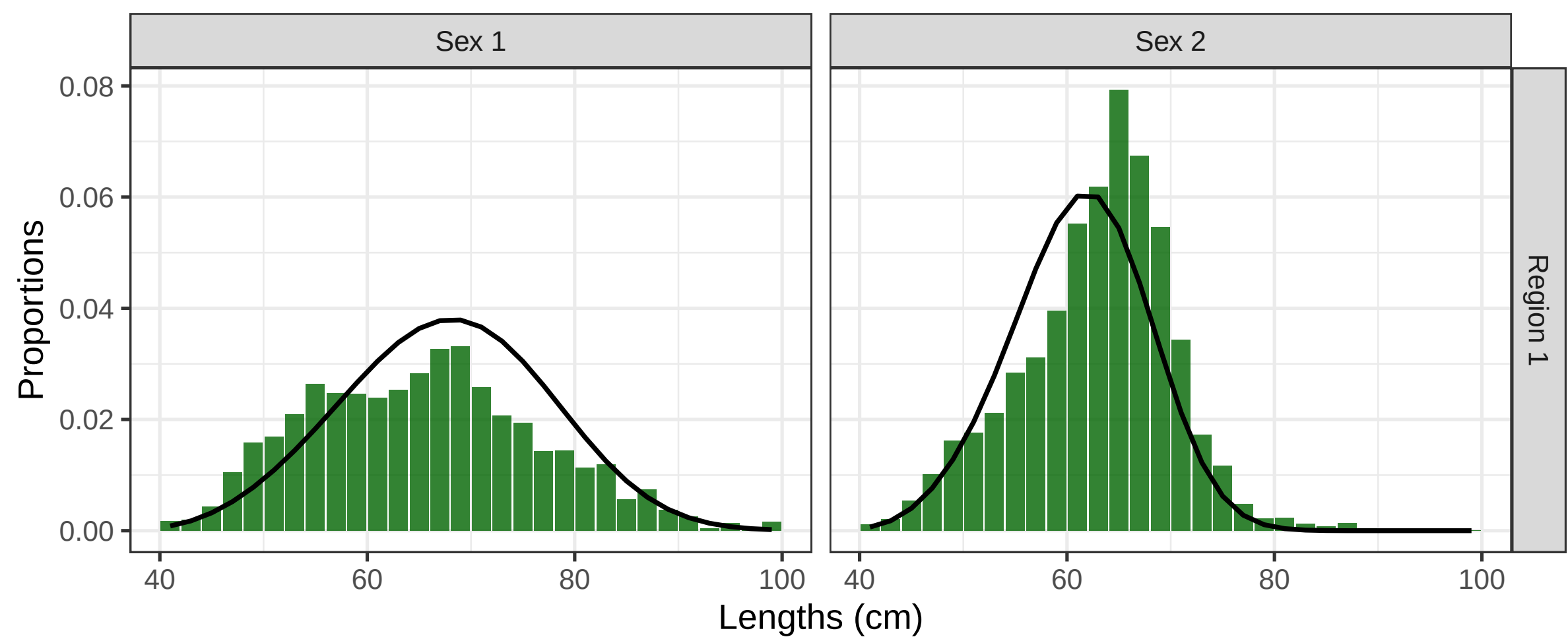
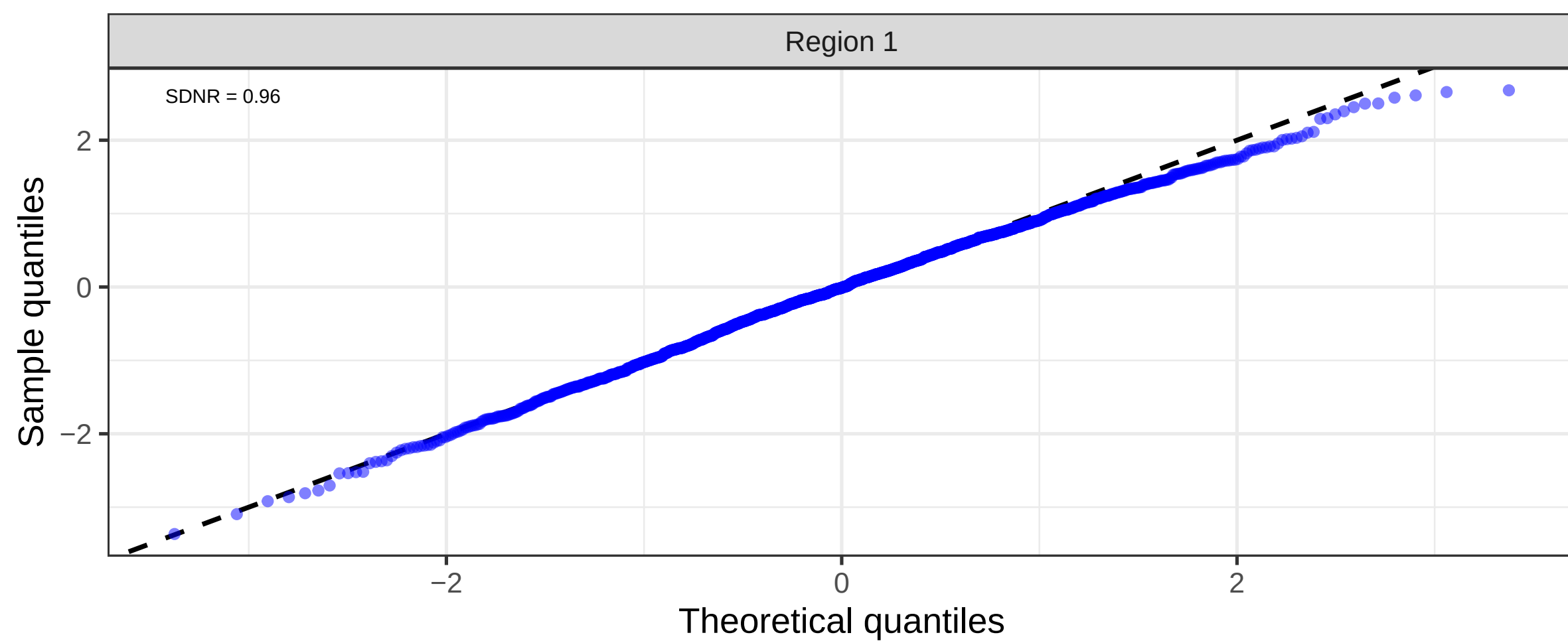
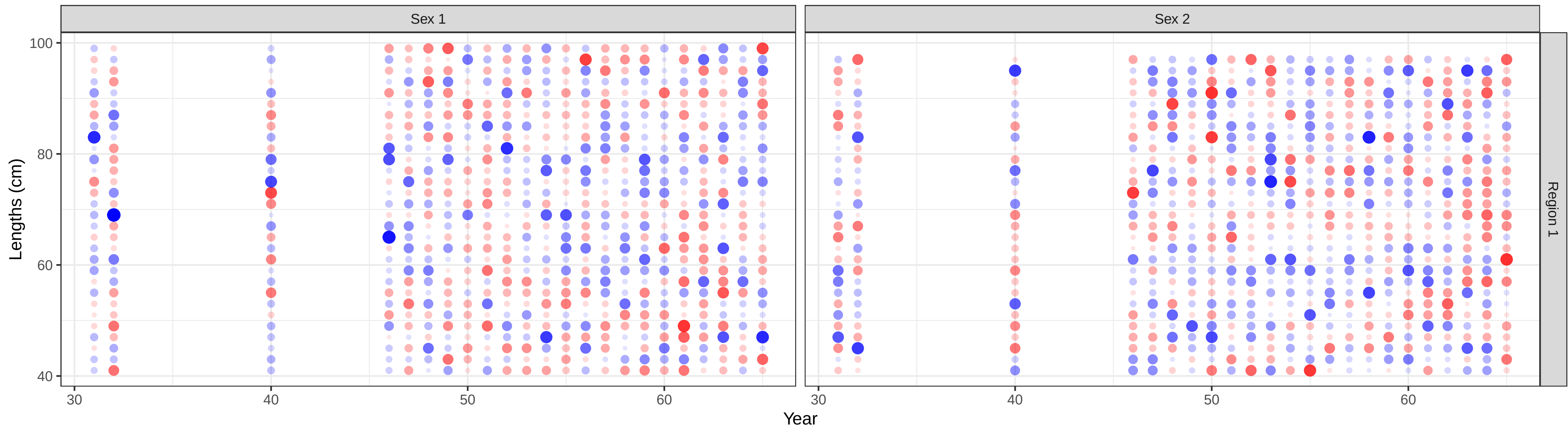
JPN LLS Length Comps Male



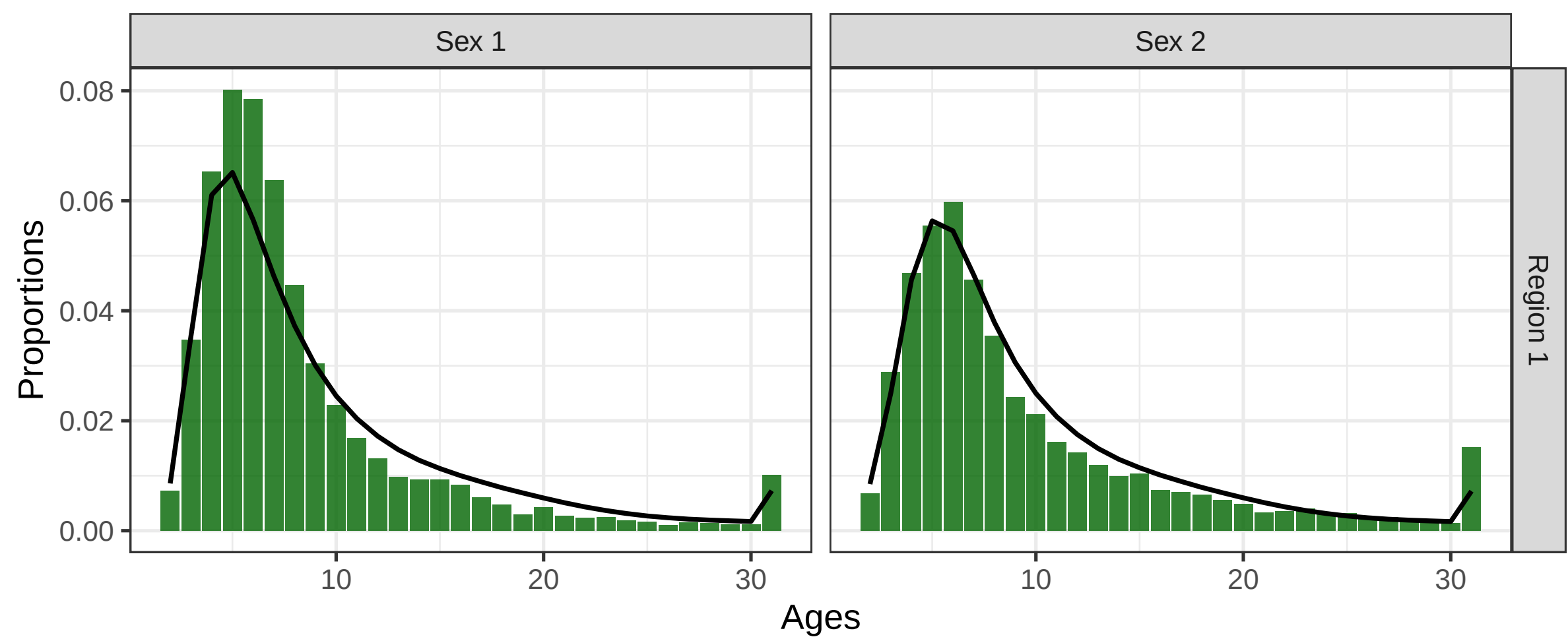
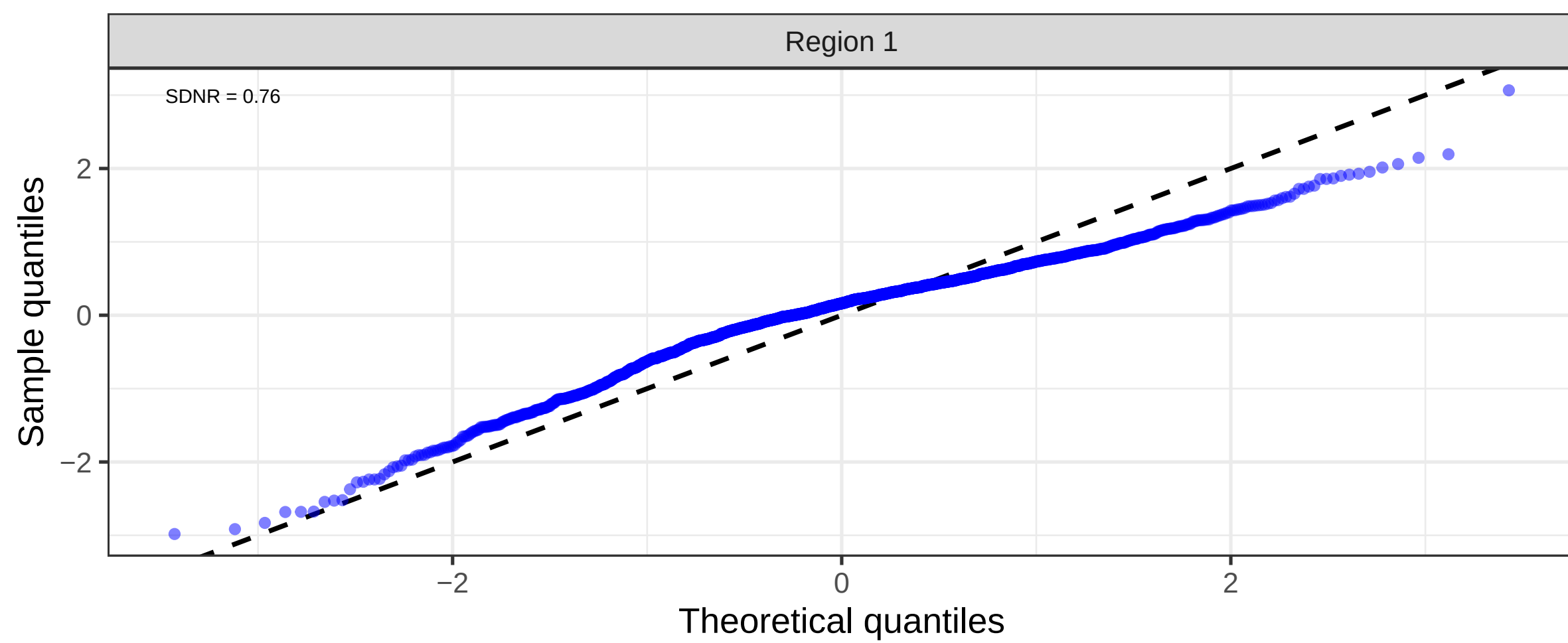
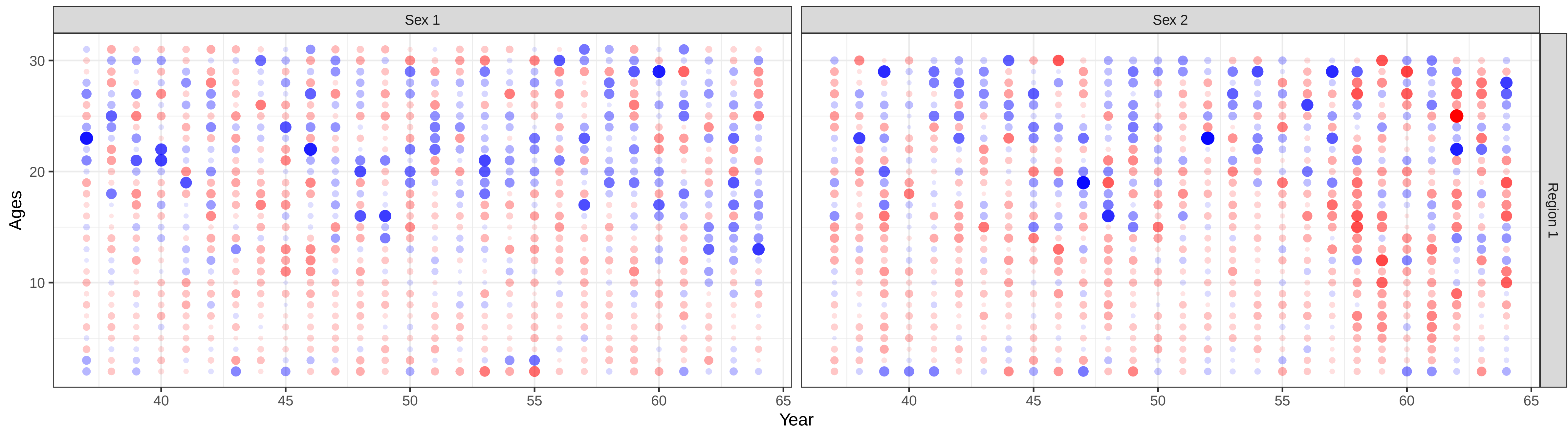
abs(Resid) 1 2 3 Sign ● Neg ● Pos

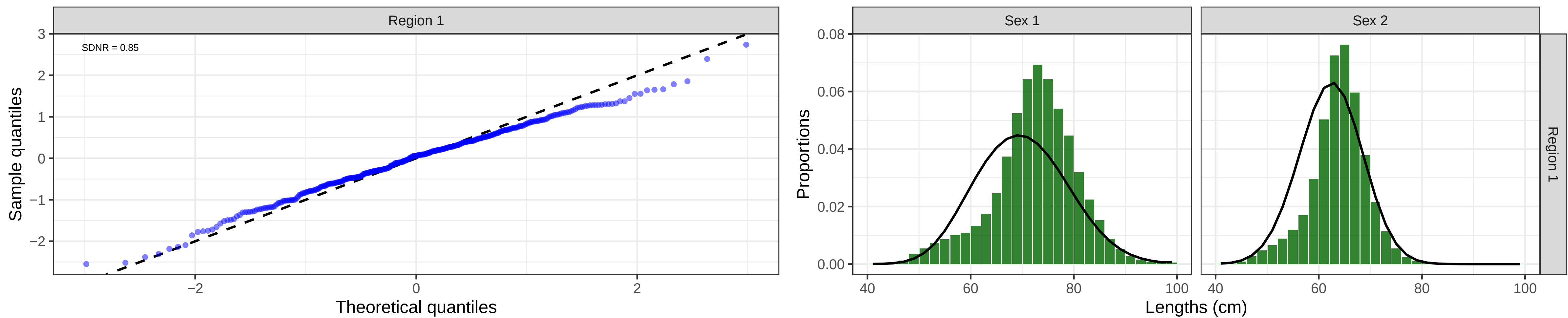
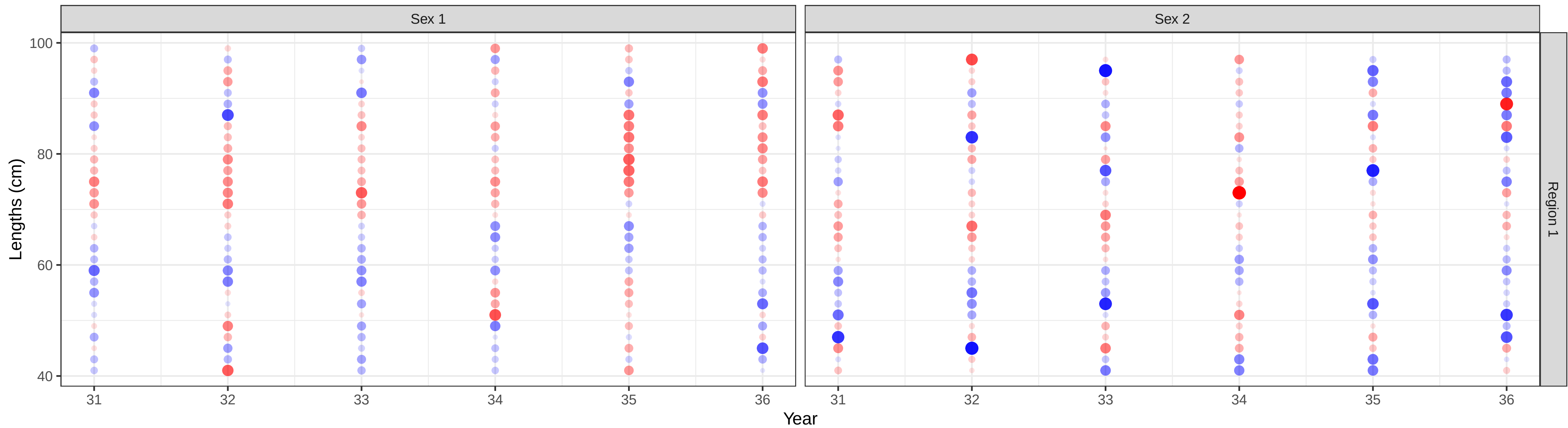




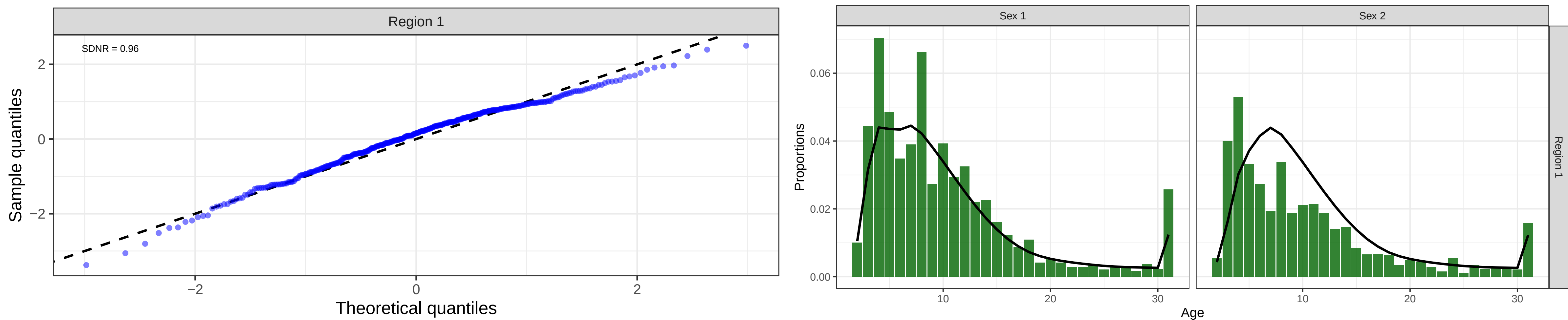
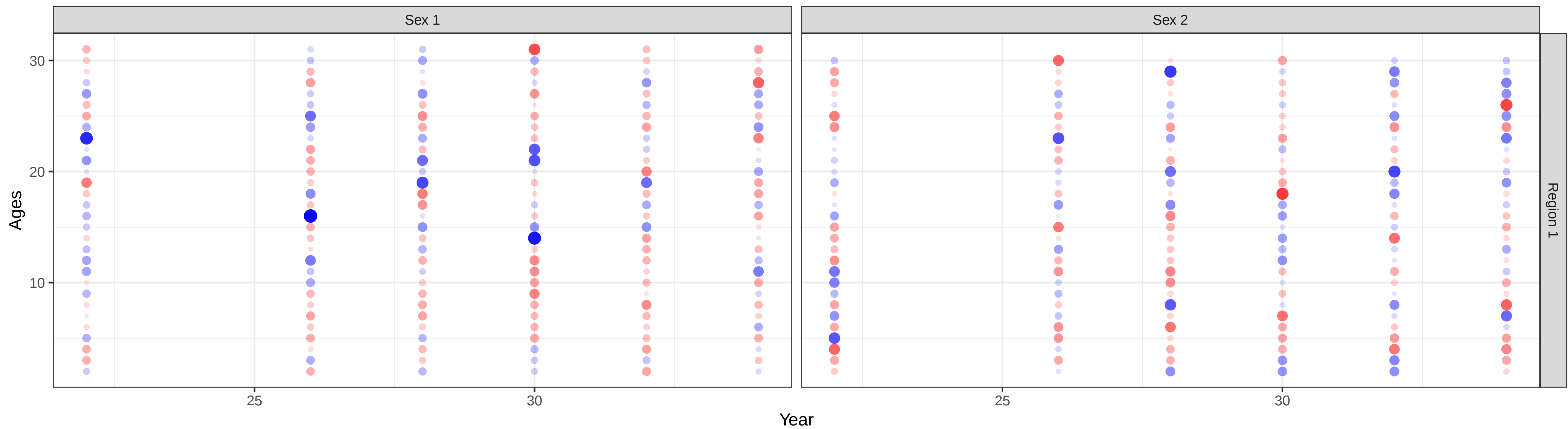


abs(Resid) 1 2 3 Sign ● Neg ● Pos





abs(Resid) 1 2 3 Sign ● Neg ● Pos



Sign ● Neg ● Pos abs(Resid) ● 1 ● 2

